■ প্রশ্ন কমনের প্রমাণ পত্র ■

প্রমাণপত্র : ২০২৪ সালের বাটাছিকম পুরাতন সাজেশন্স থেকে কমনকৃত প্রশ্নের হার ৯৮%।

National University

[BBA Hon's 3rd Year; Exam-2024 (Held on- 07/01/2026)]

(Department of Marketing)

Subject Code : 232307

Subject Name : Business Statistics-II (In English)

Time : 4 hours; **Full marks :** 80

[**N.B.** Different parts of each question must be written consecutively.]

Part-A

1. Answer any **ten** questions : $1 \times 10 = 10$

(a) Define joint probability. *[অধ্যায়-০১ এর ০৬ নং প্রশ্ন কমন পড়েছে]*

(b) What is continuous random variable? *[অধ্যায়-০১ এর ০৯৬ নং প্রশ্ন কমন পড়েছে]*

(c) What is t-distribution? *[অধ্যায়-০৫ এর ০৫ নং প্রশ্ন কমন পড়েছে]*

(d) What is contingency table? *[অধ্যায়-০৫ এর ০৪ নং প্রশ্ন কমন পড়েছে]*

(e) Define Chi-square distribution. *[অধ্যায়-০৭ এর ০৬ নং প্রশ্ন কমন পড়েছে]*

(f) What is an experiment? *[অধ্যায়-০১ এর ১২ নং প্রশ্ন কমন পড়েছে]*

(g) What is the probability distribution? *[অধ্যায়-০২ এর ০৭ নং প্রশ্ন কমন পড়েছে]*

(h) What is parameter? *[অধ্যায়-০৩ এর ০৬ নং প্রশ্ন কমন পড়েছে]*

(i) What is degree of freedom? *[অধ্যায়-০৩ এর ১৫ নং প্রশ্ন কমন পড়েছে]*

(j) What is null hypothesis? *[অধ্যায়-০৫ এর ০৩ নং প্রশ্ন কমন পড়েছে]*

(k) What is the full meaning of ANOVA? *[অধ্যায়-০৮ এর ১১ নং প্রশ্ন কমন পড়েছে]*

(l) What is multiple regression? *[অধ্যায়-০৯ এর ০৭ নং প্রশ্ন কমন পড়েছে]*

Part-B

[Answer any **five** questions.] Marks : $4 \times 5 = 20$

2. What are the properties of binomial distribution? *[অধ্যায়-০১ এর ০৬ নং প্রশ্ন কমন পড়েছে]*
3. *[অধ্যায়-০৩ এর Problem-03 নং (হুবহু) প্রশ্ন কমন পড়েছে]*
4. *[অধ্যায়-০২ এর Problem-05 নং (অনুরূপ) প্রশ্ন কমন পড়েছে]*
5. *[অধ্যায়-০৯ এর Problem-03 নং (অনুরূপ) প্রশ্ন কমন পড়েছে]*
6. *[অধ্যায়-০১ এর Problem-03 নং (হুবহু) প্রশ্ন কমন পড়েছে]*
7. Distinguish between sampling and non sampling error. *[অধ্যায়-০৩ এর ০৯ নং প্রশ্ন কমন পড়েছে]*
8. *[অধ্যায়-০২ এর Problem-01 নং (হুবহু) প্রশ্ন কমন পড়েছে]*
9. Distinguish between z-test and t-test. *[অধ্যায়-০৫ এর ০৮ নং প্রশ্ন কমন পড়েছে]*

Part-C

[Answer any **five** questions.] Marks : $10 \times 5 = 50$

10. Describe the sampling methods. *[অধ্যায়-০৩ এর ০৬ নং প্রশ্ন কমন পড়েছে]*
11. *[অধ্যায়-০২ এর Problem-08 নং (হুবহু) প্রশ্ন কমন পড়েছে]*
12. *[অধ্যায়-০৫ এর Problem-06 নং (অনুরূপ) প্রশ্ন কমন পড়েছে]*
13. *[অধ্যায়-০৮ এর Problem-05 নং (অনুরূপ) প্রশ্ন কমন পড়েছে]*
14. *[অধ্যায়-০২ এর Problem-09 নং (হুবহু) প্রশ্ন কমন পড়েছে]*
15. *[অধ্যায়-০৩ এর Problem-08 নং (হুবহু) প্রশ্ন কমন পড়েছে]*
16. *[অধ্যায়-০৫ এর Problem-12 নং (হুবহু) প্রশ্ন কমন পড়েছে]*
17. *[অধ্যায়-০৮ এর Problem-03 নং (হুবহু) প্রশ্ন কমন পড়েছে]*

National University

[BBA (Hon's) 3rd Year; Exam-2025 (Will be held-2026/27)]

Subject Code : 232307

Subject Name : Business Statistics-II

[Baticrom Latest Final Suggestions]

মার্কেটিং বিভাগের ছাত্রছাত্রীদের জন্য

Chapter 01 Probability

(সম্ভাবনা)

Part-A : Brief Questions & Solutions :

01 **What is probability?** 99%

Or, Define the term probability.

[Test Exam-Dhaka, Eden Girls (Acc.), New Model, Ananda Mohan, B.L College; N.U. BBA 2019, 2021, 2023; [D.U. (Affiliated) (Hon's) 3rd 2016 (Marketing)]

Ans : Probability is the favorable specific ratio of occurring or not occurring of any event.

[রকেট স্পেশাল সাজেশন্স : উপরিউক্ত ৯৯% প্রশ্নটি ১০০% গুরুত্বসহকারে পড়তে হবে। -সম্পাদক]

02. **What do you mean by mutually exclusive events?**

[N.U.BBA (All Dept.) 2011, (Acc)-2nd 2016]

Ans : Events that can not happen together are called mutually exclusive event.

03. **What do you mean by r = 0?**

Ans : The value of r = 0 is called zero (0) correlation. *[D. U. (2nd) 2017(Acc)]*

04. **What is meant by sample space.**

[Test Exam-Dhaka City, B.L College; N.U.BBA (All Dept.) 2011]

Ans : The set of all possible outcomes of an experiment is called the sample space.

05. **Write the formula of the addition rule for not mutually exclusive events.**

Ans : $P(A \cup B) = P(A) + P(B) - P(A \cap B)$

06. Define joint probability. [২০২৪ সালে যা এসেছিল]

[N.U.BBA (All Dept.) 2013, (2nd)-2017; 2024 (Marketing); D.U. (Affiliated) 2017(Acc)]

Ans : The probability of two events occurring together or in succession is called joint probability.

07 **What is normal curve?** 99%

[N.U. (Hon's) 3rd 2017, 2023 (Marketing)]

Ans : The probability graph of a normal distribution is called the normal curve.

[রকেট স্পেশাল সাজেশন্স : উপরিউক্ত ৯৯% প্রশ্নটি ১০০% গুরুত্বসহকারে পড়তে হবে। -সম্পাদক]

08 **What is random variable?** 99%

[N.U.BBA (All Dept.) 2014, (Acc)-2nd 2017; N.U.BBA (Finance)-2018; (Hon's) 3rd 2016, 2018, 2020, 2023 (Marketing)]

Ans : A random variable is a variable whose possible values are outcomes of a random phenomenon.

[রকেট স্পেশাল সাজেশন্স : উপরিউক্ত ৯৯% প্রশ্নটি ১০০% গুরুত্বসহকারে পড়তে হবে। -সম্পাদক]

09. What is continuous random variable? [২০২৪ সালে যা এসেছিল] [N.U.2024 (Marketing)]

Ans : A variable that is allowed to take on any value within a given range is called a continuous random variable.

10. **What Is conditional probability?**

[Test Exam-Dhaka commerce College; N.U.BBA (2nd)- 2015, 2018(Acc)]

Ans : If two events A and B are dependent then the probability is known as conditional probability.

11. **What is significance level?** 99%

[N.U.BBA 2nd 2017, 2018(Acc)]

Ans : The significance level is the probability of rejecting the null hypothesis when it is actually true (type I error) and it is most often set at 0.5 (5%)

12. What is an experiment? [২০২৪ সালে যা এসেছিল]

[N.U. (Hon's) 3rd 2017, 2019, 2021, 2024 (Marketing)]

Ans : The activity that results in, or produces and event is regarded as an experiment.

o 13. **What is venn diagram?**

Ans : A pictorial representation of probability concepts in which the sample space is represented as a rectangle and the events in the sample space as portions of that rectangle.

ooo 14. **What is the range of 0, – 2?**

[N.U. (2nd)-2018 (Acc); D.U. (Affiliated) (2nd)-2017(Acc)]

Ans : Range = Highest value – lowest value = 0 – (– 2) = 0 + 2 = 2

15. What is business statistics? 99%

[N.U. (Hon's) 3rd 2022 (Marketing)]

Ans : Business statistics is a science assisting us to make business decision under uncertainties based on some numerical and measurable scale.

[রকেট স্পেশাল সাজেশন্স : উপরিউক্ত ৯৯% প্রশ্নটি ১০০% গুরুত্বসহকারে পড়তে হবে। -সম্পাদক]

16. What is the meaning of P(A) = 1? 99%

[N.U. (Hon's) 3rd 2022 (Marketing)]

Ans : Meaning of P(A) = 1, the total number of elements of event A and sample space are equal. That is implies A is a certain event.

[রকেট স্পেশাল সাজেশন্স : উপরিউক্ত ৯৯% প্রশ্নটি ১০০% গুরুত্বসহকারে পড়তে হবে। -সম্পাদক]

Part- B & C: Broad Questions :

ooo 01. What do you mean by probability? **[Ans** মার্কেটিং প্রসঙ্গ (২য় খণ্ড) বইয়ের **306 Page 1.01 No.]**

[N.U. 2004, 2007, 2010 (old); BBA (2nd) 2018 (Acc)]

ooo 02. State the addition and multiplication theory of probability. **[Ans : 310 Page 1.20+1.21 No.]**

[Test Exam : MM College; N.U.BBA (old) 2004, 2008, 2010 2013; (Acc) 2nd 2015; (Finance-2nd) 2017; D.U. (Affiliated) (Hon's) 3rd 2016 (Marketing)]

Or, Explain the addition and multiplication theorem of probability.

oo 03. Show the distinction between continuous variable and discrete variable. **[Ans : 310 Page 1.23 No.]**

[N.U. (Old) 2008; BBA Hon's-2nd (Finance)-2018]

ooo 04. **What are the characteristics of normal distribution? Explain.** 99% **[Ans : Page No-308 Ques. No-1.13]**

[N.U. (old) 2005, 2008; Hon's 3rd 2018, 2020 (Marketing); D.U. (Affiliated) 2016 (Marketing)]

Or, Write down the properties of normal distribution.

05. Discuss the importance of probability in making business decision. 99% **[Ans : 311 Page 1.26 No.]**

[Test Exam :B.M. Edward, Chuadanga, Tularam college; N.U. 3rd 2017, 2019, 2021 (Marketing)]

Or, Discuss the importance of probability in business decision.

[রকেট স্পেশাল সাজেশন্স : উপরিউক্ত ৯৯% প্রশ্নটি ১০০% গুরুত্ব সহকারে পড়তে হবে। -সম্পাদক]

06. Discuss the uses of normal distribution. 99% **[Ans : Page No-312 Ques. No-1.29]** *[N.U. 2023 (Marketing)]*

[রকেট স্পেশাল সাজেশন্স : উপরিউক্ত ৯৯% প্রশ্নটি ১০০% গুরুত্ব সহকারে পড়তে হবে। -সম্পাদক]

ooo 07. **Show the differences between a prior and a posterior definitions of probability.** 99% **[Ans : Page No-313 Ques. No-1.32]** *[N.U. 2020, 2023 (Marketing)]*

Or, Write down the differences between a prior and a posterior definitions of probability.

[২০২৪ সালের পরীক্ষায় যা এসেছিল–]

2. What are the properties of binomial distribution? **[Ans : Page No. 309; Ques. No. 1.15]**

Or, Discuss the properties of binomial distribution *[N.U. 2012, 2020, 2022, 2024 (Marketing)]*

Help Line ***Dear Student's, In above Suggestions In Part-B and C- 05 & 06 No. Questions are most Important for the Examinee of 2025.*** ***..... Editor***

গাণিতিক সমস্যাবলি ও সমাধান (Mathematical Problems & Solution)

Easy Math Basic Problem : Category-B & C

○○ **Problem-01** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 3rd (Marketing) 2018 No–2**

Find the probability of getting 2 heads in tossing 5 coins.

Solution : মার্কেটিং এক্সপ্রেস (২য় খণ্ড) বইয়ের ৩১৪ পৃষ্ঠার **Problem-01** নং প্রশ্নের উত্তর দেখুন।

○○○ **Problem-02** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 3rd (Marketing) 2016 No–03; 2020 No–04**

Find the probability of getting 3 heads in tossing 5 coins.

Solution : ৩১৪ পৃষ্ঠার **Problem-02** নং প্রশ্নের উত্তর দেখুন।

[২০২৪ সালের পরীক্ষায় যা এসেছিল-

☒☒ **Problem-03** ***Year Question & Solution*** ⊙ **N.U. BBA (Hons.) 3rd Marketing 2017 No–3; 2021 No–7; 2024 No–6**

An unbiased coin is tossed four times. What is the probability of getting at least 2 heads?

Solution : ৩১৫ পৃষ্ঠার **Problem-03** নং প্রশ্নের উত্তর দেখুন।

○○○ **Problem-04** ***Year Question & Solution*** ⊙ **N.U. BBA (Hons.) 3rd Marketing 2022 No–12**

Three unbaised coins are tossed at a time.

(a) Write down the sample space.

(b) Calculate the probabilities of the following :

(i) 1 head

(ii) 1 tail

(iii) More than 1 head

(iv) At best 2 heads

(v) At least 2 heads.

Solution : ৩১৫ পৃষ্ঠার **Problem-04** নং প্রশ্নের উত্তর দেখুন।

[রকেট স্পেশাল সাজেশন্স : উপরিউক্ত প্রশ্নটি ১০০% গুরুত্ব সহকারে পড়তে হবে। -সম্পাদক]

○○ **Problem-05** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 2nd (Old) 2013 No–17; (Acc.) 2022 No–07; 3rd (Mgt.) 2016 No–07**

A bag contains 7 red, 12 white and 4 green balls.

What is the probability that :

(i) 3 balls drawn are all white and

(ii) 3 balls drawn are one of each colour.

Solution : ৩১৮ পৃষ্ঠার **Problem-09** নং প্রশ্নের উত্তর দেখুন।

○○ **Problem-06** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 2nd (Acc.) 2015 No–14(b); (Finance) 2022 No–15(b)**

There are 5 white and 7 red balls in a bag. Two balls are drawn such that a ball is drawn and replaced. What is the probability that a white ball and a red ball are drawn in that order? What would be the probability if the balls drawn were no put back into the bag?

Solution : ৩১৮ পৃষ্ঠার **Problem-10** নং প্রশ্নের উত্তর দেখুন।

০০০ **Problem-07** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 3rd (Marketing) 2017 No–10; 2020 No–10; 2023 No– 10**

A bag contains 6 red and 4 white balls. Two balls are drawn at random, What is the probability that :

(i) All of them are red

(ii) Same colour,

(iii) Different colour

(iv) None of them is red.

Solution : ৩২০ পৃষ্ঠার **Problem-13** নং প্রশ্নের উত্তর দেখুন।

৩৫৪ (৯৯%) [রকেট স্পেশাল সাজেশন্স : উপরিউক্ত প্রশ্নটি ১০০% গুরুত্ব সহকারে পড়তে হবে। –সম্পাদক]

০০০ **Problem-08** ***Year Question & Solution*** ⊙ **N.U. BBA (Hons.) 3rd Marketing 2021 No–10**

A box contains 6 red, 7 blue and 8 green balls. If 3 balls are drawn at random, determine the probability that :

(i) All 3 are red

(ii) All 3 are blue

(iii) At least 1 is blue

(iv) 2 are red and 1 is green

(v) 1 of each color.

Solution : ৩২৪ পৃষ্ঠার **Problem-16** নং প্রশ্নের উত্তর দেখুন।

০০০ **Problem-09** ***Year Question & Solution*** ⊙ **N.U. BBA (Hons.) 3rd Marketing 2022 No–2**

A card is drawn from a pack of 52 cards. Find the probability of getting (i) King (ii) Club.

Solution : ৩২৬ পৃষ্ঠার **Problem-18** নং প্রশ্নের উত্তর দেখুন।

০০ **Problem-10** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 2nd (Old) 2012 No–5; (Management) 3rd 2018 No–7**

4 cards are drawn from a pack of 52 cards without replacement. Find the probability of getting 3 aces.

Solution : ৩২৬ পৃষ্ঠার **Problem-20** নং প্রশ্নের উত্তর দেখুন।

০০০ **Problem-11** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 3rd (Marketing) 2019 No–13; 2021 No–15**

Two cards are drawn without replacement from a pack of 52 cards. Find the probability that the two cards are :

(i) Same color Ace

(ii) No king

(iii) Different colors.

Solution : ৩২৭ পৃষ্ঠার **Problem-21** নং প্রশ্নের উত্তর দেখুন।

০০ **Problem-12** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 2nd (Finance) 2018 No–14**

Four card are drawn at random from a pack of 52 cards. Find the probability that :

(a) They are a king, a queen, a jack and an ace.

(b) Two are kings and two are aces.

(c) All are diamonds.

(d) Two are red and two are black.

(e) There is one card of each suit,

(f) There are two cards of clubs and two cards of diamonds.

Solution : ৩২৮ পৃষ্ঠার **Problem-22** নং প্রশ্নের উত্তর দেখুন।

Chapter 02 Probability Distributions

(সম্ভাবনা বিন্যাস)

Part-A : Brief Questions & Solutions:

01. What is binomial distribution? 99%

[N.U. (Hon's) 3rd 2019, 2021 (Marketing)]

Ans : The distribution which contains only two possible outcomes (such as success or failure) is called binomial distribution.

[রকেট স্পেশাল সাজেশন্স : উপরিউক্ত ৯৯% প্রশ্নটি ১০০% গুরুত্বসহকারে পড়তে হবে। -সম্পাদক]

02. What is normal distribution? 99%

[N.U. (Hon's) 3rd 2017, 2019 (Marketing);D.U. (Affiliated) (Hon's) 3rd 2016 (Marketing)]

Or, What is standard normal distribution?

Ans : A distribution of a continuous random variable with a single peaked, bell-shaped curve is called a standard normal distribution.

[রকেট স্পেশাল সাজেশন্স : উপরিউক্ত ৯৯% প্রশ্নটি ১০০% গুরুত্বসহকারে পড়তে হবে। -সম্পাদক]

03. What is conditional probability?

[Test Exam-Dhaka commerce College N.U.BBA -2015]

Ans : If two events A and B are dependent then the probability is known as conditional probability.

04. What are equally likely event?

Ans : The outcomes of a trial or events are said to be equally likely if each of them have equal chance to be occurred, that is when one does not occur more after the others

05. What is event? 99%

[N.U. (Hon's) 3rd 2016, 2018, 2020 (Marketing)]

Ans : One or more of the possible outcomes of doing something, or one of the possible outcomes from conducting an experiment.

06. What is probability tree?

Ans : Probability tree is a graphical representation showing the possible outcomes of a series of experiments and their respective probabilities.

07. What is the probability distribution? [২০২৪ সালে যা এসেছিল] [N.U.BBA -2016, 2020, 2024]

Ans : Probability distribution is a list of all possible outcomes of an experiment and the probabilities associated with each outcome.

08. What is F- distribution?

[D.U. (Affiliated) (Hon's) 3rd 2016 (Marketing)]

Ans : F- distribution is a test statistic used to test the hypothesis that population variance are equal. It is used to analysis of variance.

09. What is Poisson distribution? 99%

[Test Exam-Mohammadpur Kendriy; N.U. (Hon's) 3rd 2023 (Marketing)]

Ans : A Poisson distribution is a discrete probability distribution in which the probability of occurrence of an outcome wit in very small time interval is small and the probability of two or more such outcomes will occur within the same interval is negligible. The occurrence of an outcome within one time period is independent of the other.

[রকেট স্পেশাল সাজেশন্স : উপরিউক্ত ৯৯% প্রশ্নটি ১০০% গুরুত্বসহকারে পড়তে হবে। -সম্পাদক]

10. What is the formula of poisson distribution? 99%

[N.U. (Hon's) 3rd 2022 (Marketing)]

Ans : The formula of poisson distribution

$$P(x) = \frac{e^{-m} m^x}{x!}$$

Part- B & C: Broad Questions :

01. What is probability distribution? 99% **[Ans : অ্যাডভান্সড মার্কেটিং ম্যানেজমেন্ট (২য় খণ্ড) বইয়ের 307 Page 1.05 No.]**

Or, Define the probability distribution. *[Test Exam- Shah Sultan, Nator, MM College]*

02. Write down the importance of poisson distribution? **[Ans : 309 Page 1.19 No.]**

Or, Write down the applications of poisson distribution? *[Test Exam-Mirpur, Edward Satkhira College]*

03. Explain the concept of conditional probability. *[N.U. (Hon's) 3rd 2016, 2018 (Marketing)]*

04. What are the characteristics of Poisson distribution? 99% **[Ans : Page No-309 Ques. No-1.17]**

[N.U. 3rd 2019, 2021, 2023 (Marketing)]

[রকেট স্পেশাল সাজেশন্স : উপরিউক্ত ৯৯% প্রশ্নটি ১০০% গুরুত্ব সহকারে পড়তে হবে। -সম্পাদক]

Help Line ***Dear Student's, In above Suggestions in Part-B and C- 04 No. Questions are most important for the Examinee of 2025.*** ***..... Editor***

গাণিতিক সমস্যাবলি ও সমাধান (Mathematical Problems & Solution)

Easy Math Basic Problem : Category-B & C

○○○ **Problem-01** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 3rd (Marketing) 2020 No–08; 2023 No– 8**

The mean and variance of a binomial distribution are 4 and $\frac{4}{3}$ respectively.

Find (i) probability function (ii) P (x = 0), (iii) P (x ≥ 1).

Solution : বেস্ট অফ মার্কেটিং হাইলাইটস (২য় খণ্ড) বইয়ের ৩৩৪ পৃষ্ঠার **Problem-07** নং প্রশ্নের উত্তর দেখুন।

○○○ (৯৯%) [রকেট স্পেশাল সাজেশন্স : উপরিউক্ত প্রশ্নটি ১০০% গুরুত্ব সহকারে পড়তে হবে। –সম্পাদক]

[২০২৪ সালের পরীক্ষায় যা এসেছিল-

☒☒ **Problem-02** ***Year Question & Solution*** ⊙ **N.U. BBA (Hons.) 2nd (Acc.) 2017 No–8; 3rd (Mgt) 2018 No–6; 3rd (Marketing) 2018 No–4; 2022 No–7; 2024 No–8**

The mean of binomial distribution is 40 and standard deviation is 6. Calculate N, P and Q.

Or, The mean of a binomial distribution is 40 and standard deviation is 6. Calculate the values of n, p and q.

Solution : ৩৩২ পৃষ্ঠার **Problem-04** নং প্রশ্নের উত্তর দেখুন।

○○○ **Problem-03** ***Year Question & Solution*** ⊙ **N.U. BBA (Hons.) 3rd Marketing 2022 No–5**

Assume that on an average one telephone number out of 15 is busy. What is the probability that if six randomly selected telephone numbers are called not more than three will be busy?

Solution : ৩৩৫ পৃষ্ঠার **Problem-08** নং প্রশ্নের উত্তর দেখুন।

○○○ **Problem-04** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 3rd (Marketing) 2018 No–11; 2021 No–11; 2023 No– 11**

The probability that a patient survives from neurosurgery is 0.70. If 10 patients have had the surgery, then what is the probability that :

(i) None will survive.

(ii) At least one will survive.

(iii) At most three will survive.

(iv) Between two and four will survive.

Solution : ৩৩৬ পৃষ্ঠার **Problem-09** নং প্রশ্নের উত্তর দেখুন।

○○○ (৯৯%) [রকেট স্পেশাল সাজেশন্স : উপরিউক্ত প্রশ্নটি ১০০% গুরুত্ব সহকারে পড়তে হবে। –সম্পাদক]

○○ **Problem-05** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 2nd (Finance) 2022 No–7; 3rd (Mgt) 2017 No–13; 2019 No–14**

About 20% items of a production are defective in a certain factory. 10 products are packed in a packet. Among the 2000 packets find the probability of getting packet containing :

(i) No defective product;

(ii) At least one product is defeotive;

(iii) At best one product is defective.

Solution : ৩৩৬ পৃষ্ঠার **Problem-10** নং প্রশ্নের উত্তর দেখুন।

[২০২৪ সালের পরীক্ষায় যা এসেছিল-

ⓍⓍ **Problem-06** ***Year Question & Solution*** ⊙ **N.U. BBA (Hons.) 3rd Marketing 2024 No–4**

The probability of a man to be smoker is $\frac{1}{3}$. What is the probability of 2 men among 5 men to be smokers?

Solution : ৫৪৪ পৃষ্ঠার **Question No-4** নং প্রশ্নের উত্তর দেখুন।

ooo **Problem-07** ***Year Question & Solution*** ⊙ **N.U. BBA (Hons.) 3rd Marketing 2022 No–11**

In a community the probability that a newly born child will be a girl is 0.4. Among 4 newly born children in that community, determine the probability that :

(i) All the four girls
(ii) At least two girls
(iii) No girl
(iv) Exactly one girl
(v) At most two girls.

Solution : ৩৩৭ পৃষ্ঠার **Problem-11** নং প্রশ্নের উত্তর দেখুন।

ooo **Problem-08** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 3rd (Marketing) 2023 No– 4**

In a Poisson distribution, $P(x = 2) = P(x = 3)$. Find the probability function.

Solution : ৩৪০ পৃষ্ঠার **Problem-14** নং প্রশ্নের উত্তর দেখুন।

ooo (৯৯%) [রকেট স্পেশাল সাজেশন্স : উপরিউক্ত প্রশ্নটি ১০০% গুরুত্ব সহকারে পড়তে হবে। –সম্পাদক]

[২০২৪ সালের পরীক্ষায় যা এসেছিল-

ⓍⓍ **Problem-09** ***Year Question & Solution*** ⊙ **N.U. BBA (Hons.) 2nd (Old) 2014 No–17; (Acc.) 2017 No–17; 2nd (Finance) 2019 No–17(b); (Mgt) 3rd 2018 No–15(b); 3rd Marketing 2024 No–11**

A telephone operator receives 3 telephone calls on average per minute from 10 a.m. to 12 p.m. Find the probability that in a certain minute of the given time interval of a day, the operator receives :

(i) No call
(ii) At least two calls
(iii) At best two calls
(iv) Two or three calls.

Solution : ৩৪১ পৃষ্ঠার **Problem-16** নং প্রশ্নের উত্তর দেখুন।

[২০২৪ সালের পরীক্ষায় যা এসেছিল-

ⓍⓍ **Problem-10** ***Year Question & Solution*** ⊙ **N.U. BBA (Hons.) 3rd Marketing 2020 No–12; 2022 No–10; 2024 No–14**

On average, one in 400 items is defective. If the items are packed in boxes of 100. What is the probability that any given box of items will contain :

(i) No defective
(ii) Less than two defectives
(iii) One or more defectives
(iv) More than three defectives.

Solution : ৩৪২ পৃষ্ঠার **Problem-17** নং প্রশ্নের উত্তর দেখুন।

০০০ **Problem-11** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 3rd (Marketing) 2016 No–11; 2023 No– 12**

The amount of money requested on loan applications at 'Pubali Bank'/'ABC Bank' follow the normal distribution, with a mean of Tk. 70,000 and standard deviation of Tk. 20,000. A loan application is recorded this morning. Find the probability that :

(i) The amount requested is Tk. 80,000 or more.

(ii) The amount requested is between Tk. 65,000 and Tk. 80,000.

Solution : ৩৪৫ পৃষ্ঠার **Problem-19** নং প্রশ্নের উত্তর দেখুন।

০০০ (৯৯%) [রকেট স্পেশাল সাজেশন্স : উপরিউক্ত প্রশ্নটি ১০০% গুরুত্ব সহকারে পড়তে হবে। –সম্পাদক]

০০০ **Problem-12** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 3rd (Marketing) 2020 No–11; 2022 No–13**

The income of a group of 10000 persons was found to be normally distributed with mean Tk. 1750 p.m. and standard deviation Tk. 50. Show that of this group 95% had income exceeding Tk. 1668 and only 5% had income exceeding Tk. 1832. What was the lowest income among the richest 100%.

Solution : ৩৪৬ পৃষ্ঠার **Problem-20** নং প্রশ্নের উত্তর দেখুন।

০০০ **Problem-13** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 3rd (Marketing) 2017 No–12**

The average daily sales of 500 baranch offices was Rs. 150 thousand and the standard deviation Rs. 15 Thousand. Assuming the distribution to be normal, indicate how many branches have sales between.

(i) Rs. 120 thousand and Rs. 145 thousand.

(ii) Rs. 140 thousand and 165 thousand.

Solution : ৩৪৮ পৃষ্ঠার **Problem-22** নং প্রশ্নের উত্তর দেখুন।

০০ **Problem-14** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 2nd (Accounting) 2017 No–15**

An industry has 10 male and 7 female workers. 5 workers selected at random. Find the probability of getting 3 female workers at least.

Solution : ৩৪৯ পৃষ্ঠার **Problem-24** নং প্রশ্নের উত্তর দেখুন।

০০০ **Problem-15** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 3rd (Marketing) 2018 No–8; 2022 No–6**

In a survey of 100 readers, it was found 40 read the Daily Ittefaq and 15 read the Daily Star and 10 read both. What is the probability of a person reading at least one of the newspapers?

Solution : ৩৫০ পৃষ্ঠার **Problem-26** নং প্রশ্নের উত্তর দেখুন।

০০০ (৯৯%) [রকেট স্পেশাল সাজেশন্স : উপরিউক্ত প্রশ্নটি ১০০% গুরুত্ব সহকারে পড়তে হবে। –সম্পাদক]

০০০ **Problem-16** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 2nd (Old) 2011 No–13; N.U. (Management) 3rd 2016 No–13; D.U. (Finance) 2017 No–15**

A manufacturing plant produces steel pipes in three plants with daily production volumes of 500, 1000 and 2000 units respectively. According to past experience, it is known that the fractions of defective output produced by the three plants are respectively at random 0.005,0.008 and 0.010. If a pipe is selected from a day's total production and found to be defective, find out (i) from which plant for this defective pipe, the probability is highest and (ii) what is the probability that it came from the first plant?

Solution : ৩৫১ পৃষ্ঠার **Problem-27** নং প্রশ্নের উত্তর দেখুন।

oo **Problem-17** ***Year Question & Solution*** ⊙ **D.U. BBA (Hon's) 3rd (Marketing) 2016 No–15**

As a result of tests on 20,000 electric fans manufactured by a company, it was found that life time of the fans was normally distributed with an average life of 2040 hours and standard deviation of 60 hours. On the basis of the information, estimate the number of fans that is expected to run for (i) more than 2150 hours and (ii) less than 1960 hours.

Solution : ৩৫২ পৃষ্ঠার **Problem-28** নং প্রশ্নের উত্তর দেখুন।

oo **Problem-18** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 3rd (Marketing) 2017 No–4**

A hawker earn Tk. 4,000, if it is fair day. If it rains, he loss Tk. 500. If the probability of fair day is 0.6, then find his expected profit or loss per day.

Solution : ৩৫৩ পৃষ্ঠার **Problem-30** নং প্রশ্নের উত্তর দেখুন।

ooo **Problem-19** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 3rd (Marketing) 2023 No– 7**

A hawker earn Tk. 2000, if it is fair day. If it rains, he losses Tk. 250. If the probability of fair day is 0.6, then find his expected profit or loss per day.

Solution : ৩৫৩ পৃষ্ঠার **Problem-31** নং প্রশ্নের উত্তর দেখুন।

ooo (৯৯%) [রকেট স্পেশাল সাজেশন্স : উপরিউক্ত প্রশ্নটি ১০০% গুরুত্ব সহকারে পড়তে হবে। -সম্পাদক]

oo **Problem-20** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 3rd (Marketing) 2017 No–11; 2019 No–4**

You are given the following probability distribution for the annual sales for a company :

Sales (Tk. in crore)	Probability
200	0.10
250	0.20
275	0.40
310	0.25
350	0.50

Calculate the expected sale of the company. Also find the standard deviation of the sale.

Solution : ৩৫৩ পৃষ্ঠার **Problem-32** নং প্রশ্নের উত্তর দেখুন।

Chapter 03 Sampling & Sampling Distribution

(নমুনায়ন এবং নমুনায়ন বিন্যাস)

Part-A : Brief Questions & Solutions :

ooo (01) **What is sampling? 99%**

[N. U.2nd (Acc)- 2nd 2018; (Hon's) 3rd 2017, 2019, 2021, 2023 (Marketing)]

Ans : The methods of selecting a sample from a population is known as sampling.

[রকেট স্পেশাল সাজেশন্স : উপরিউক্ত ৯৯% প্রশ্নটি ১০০% গুরুত্বসহকারে পড়তে হবে। -সম্পাদক]

ooo (02) **What is population? 99%**

[N.U.BBA (All) 2011; (Hon's) 3rd 2017, 2019, 2021, 2023 (Marketing)]

Ans : A population is the collection of all possible objects having certain characteristic under a study.

[রকেট স্পেশাল সাজেশন্স : উপরিউক্ত ৯৯% প্রশ্নটি ১০০% গুরুত্বসহকারে পড়তে হবে। -সম্পাদক]

ooo 03. **What is sample?** 99% *[N.U.BBA (Finance)-2018; (Hon's) 3rd 2018, 2020 (Marketing)]*

Ans : Sample is a portion of the elements of a population selected for direct measurement.

ooo 04. **What do you mean by random sampling?**

[D.U. (Affiliated) (Hon's) 3rd 2016 (Marketing)]

Ans : The method where chance of selection for each element from a population equal is called random sampling.

oo 05. **Define finite population.**

Ans : A finite population is the aggregation of all the elements that are definite and countable under certain study.

06. ***What is parameter[২০২৪ সালে যা এসেছিল]***

[N.U.BBA (All Dept.) 2011, (Acc) 2nd 2015; (Hon's) 3rd 2017, 2020, 2022, 2024 (Marketing)]

Ans : *The values that desirable the characteristics of a population are called parameters.*

ooo 07. **What is parametric test?** 99%

[N.U. (Hon's) 3rd 2023 (Marketing)]

Ans. A parametric test is a type of statistical test that makes specific assumptions about the parameters of the population distribution from which the sample is drawn.

oo 08. **An unbiased coin is tossed three times. Write down the sample space.**

[N.U. (Hon's) 3rd 2022 (Marketing)]

Ans : Determination of sample space :

Total sample are $= 2^n = 2^3 = 8$

2 coin / 1 coin	HH	HT	TH	TT
H	HHH	HHT	HTH	HTT
T	THH	THT	TTH	TTT

∴ Sample space, s = {HHH, HHT, HTH, HTT, THH, THT, TTH, TTT}

ooo 09. **What is cluster sampling?**

[Test Exam-Dhaka College (Acc.) N.U 2013]

Ans : Cluster sampling is a method of random Sampling in which the Population is divided into groups.

oo 10. **What is sampling error?** 99%

[Test Exam-Dhaka City, Eden (Acc.), Eden College (Mgt); N.U.BBA (All Dept.) 2011; 2020, 2023 (Marketing); D.U. (Affiliated) 2nd 2017]

Ans : The error that arises due to drawing inference about a population based on Sample is known sampling error.

[রকেট স্পেশাল সাজেশন্স : উপরিউক্ত ৯৯% প্রশ্নটি ১০০% গুরুত্বসহকারে পড়তে হবে। -সম্পাদক]

ooo 11. **What is non sampling error?**

Ans : An error that may affect any sample estimate and is not related to the act of sampling.

ooo 12. **What is random sampling?**

[N.U.BBA (Acc) 2nd 2016]

Ans : The random sampling methods are those in which each and every unit in the population has an equal chance of being included in the sample.

oo 13. **Systematic sampling.** *[D. U. (Finance) 2nd 2017]*

Ans : The process of systematic sampling typically involves first selecting a fixed standing point in the larger population and then obtaining subsequent observation by using constant interval between sample taken.

o 14. **What is sampling fraction?**

Ans : The fraction of population contained in a sample is called sampling fraction.

15. ***What is degree of freedom? [২০২৪ সালে যা এসেছিল]***

[Test Exam-Govt. Bangla, Habibullahbaher College (Acc.); N.U. (Hon's) 3rd 2016, 2018, 2020, 2024 (Marketing)]

Or, What do you mean by degree of freedom?

Ans : *The number of unrestricted chances for variation in the measurement being made is called the degree of freedom.*

ooo 16. **What is meant by survey?** 99%

[N.U. (Hon's) 3rd 2018 (Marketing)]

Ans : A survey is a systematic method of collecting data from a sample of individuals or items to gather information, opinions or characteristics of a population.

ooo 17. **What is sample survey?** *[N.U.BBA (Acc)2nd 2017]*

Ans : A technique of selecting a part of the population and collecting information from the selected individuals in known as sample survey.

oo 18. **What is sampling distribution?**

Ans : A probability distribution consisting of all possible values of a sample statistic.

oo 19. **What is census survey?** 99%

[N.U. (Hon's) 3rd 2018, 2020, 2023 (Marketing)]

Or, What is census?

Ans : The measurement or examination of every element in the population is called census.

[রকেট স্পেশাল সাজেশন্স : উপরিউক্ত ৯৯% প্রশ্নটি ১০০% গুরুত্বসহকারে পড়তে হবে। -সম্পাদক]

Part- B & C: Broad Questions :

০০ 01. What is sampling? [Ans মার্কেটিং ইনসাইটস (২য় খণ্ড) বইয়ের 355 Page 3.05 No.]
Or, What do you mean by sampling? *[N.U. (old) 2004, 2006, 2007, 2009, 2010]*

০০০ 02. **State the differences between population and sample.** 99% **[Ans : 356 Page 3.08 No.]**
[Test Exam : Azizul, Sadat, Ralngur, Satkhir college; N.U.BBA (old) 2009; N.U. (Hon's) 3rd 2017 (Marketing); 2015, 2016 (Finance-2nd)]
Or, Distinction between population and sample.

০০০ 03. What are the sources of non-sampling errors? How these errors can controlled? **[Ans : 357 Page 3.12 No.]**
[Test Exam-Bhala, Burhanuddin and MM College; D.U. (Affiliated) 2016 (Markating-3rd)]

০০ 04. Difference between simple random sampling and cluster sampling. **[Ans : 360 Page 3.22 No.]**

০০ 05. What are sampling and non sampling errors? What are the causes of sampling error.
[Ans : 358 Page 3.16+3.17 No.] *[N.U.BBA (All Dept.) 2005, 2008, 2009]*

০০০ 06. Explain the importance of sampling in solving business problem.
[Ans : 360 Page 3.23 No.] *[Test Exam : Khulna, Rajshahi, B.M., Munsur, Tongi college; N.U.BBA (old) 2004]*

০০ 07. Describe the two important principles of sampling. **[Ans : 360 Page 3.24 No.]**
[N. U. (old) 2004, 2009; D.U. (Affiliated) 2017 (Finance-2nd)]

০০০ (08) **Distinguish between census and sample.** 99% **[Ans : Page No-357 Ques. No-3.15]**
[N.U. BBA (Hon's) 3rd 2017, 2019, 2021, 2023 (Marketing)]
Or, Distinguish between a sample survey and census survey.
[রকেট স্পেশাল সাজেশন্স : উপরিউক্ত ৯৯% প্রশ্নটি ১০০% গুরুত্বসহকারে পড়তে হবে। -সম্পাদক]

০০০ (09) **(a) Discuss the reasons of sample survey.** 99% **[Ans : Page No-356 Ques. No-3.06]**
Or, What are the reasons of sample survey? *[N.U.BBA 2nd 2018 (Acc); 2019, 2021, 2023 (Marketing) 3rd]*
Or, State the reasons of sample survey.
(b) Explain the ways of reducing non-sampling error. [Ans : Page No-357 Ques. No-3.14]
[N.U.BBA 2019 , 2021, 2023(Marketing) 3rd]
[রকেট স্পেশাল সাজেশন্স : উপরিউক্ত ৯৯% প্রশ্নটি ১০০% গুরুত্বসহকারে পড়তে হবে। -সম্পাদক]

[২০২৪ সালের পরীক্ষায় যা এসেছিল–

7. *Distinguish between sampling and non sampling error.* **[Ans : Page No. 358; Ques. No. 3.18]**
Or, Write down the differences between sampling and non sampling error.
[N.U. (Hon's) 3rd 2016, 2018, 2020, 2022, 2024 (Marketing)]

10. *Describe the sampling methods.* **[Ans : Page No. 359; Ques. No. 3.19]**
Or, Discuss the various methods of sampling. *[Test Exam : Comilla College; N.U.BBA -2006, 2010; 2024 (Marketing)]*
Or, Briefly discuss the various methods of sampling with suitable examples.

Help Line ***Dear Student's, In above Suggestions in Part-B and C- 02, 08 & 09 No. Questions are most important for the Examinee of 2025.*** ***..... Editor***

গাণিতিক সমস্যাবলি ও সমাধান (Mathematical Problems & Solution)

Easy Math Basic Problem : Category-B

০০ **Problem-01** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 2nd (Finance) 2018 No–6**
The mean age of a combined group of men and women is 30 years. If mean age of the group of men is 32 and that of the group of women is 25, find out the percentage of men and women in the group.

Solution : [খোঁয়াড়কেরা মার্কেটিং ইনসাইটস (২য় খণ্ড) বইয়ের ৩৬২ পৃষ্ঠার **Problem-01** নং প্রশ্নের উত্তর দেখুন।

ooo **Problem-02** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 2nd (Old) 2012 No–15; D.U. Affiliated (Acc.) 2017 No–16; (Mgt) 3rd 2016 No–15; N.U. (MBA Final) Mgt 2016 No– 07**

In a simple random sample of 600 men taken from a big city, 400 are found to be smokers. In another simple random sample of 900 men taken from another city, 450 are found to be smokers. Does the data indicate that there is a significant difference in the habit of smoking in the two cities?

Solution : ৩৬২ পৃষ্ঠার **Problem-02** নং প্রশ্নের উত্তর দেখুন।

[২০২৪ সালের পরীক্ষায় যা এসেছিল-

XX **Problem-03** ***Year Question & Solution*** ⊙ **N.U. BBA (Hons.) 3rd Marketing 2016 No–5; 2024 No–3**

A sample of 81 observations is taken from a normal population. The sample mean is 40 and the sample standard deviation is 5. Determine the 95% confidence interval for the population mean.

Solution : ৩৬৩ পৃষ্ঠার **Problem-04** নং প্রশ্নের উত্তর দেখুন।

oo **Problem-04** ***year Question & Solution*** ⊙ **N.U. BBA (Hon's) 2nd (Finance) 2018 No–16(b)**

In a sample of 1,000 persons from the village of Chatkhil, 660 are found to be consumers of rice and the rest consumers of wheat. Can it be concluded that both the food articles are equally popular?

Solution : ৩৬৩ পৃষ্ঠার **Problem-03** নং প্রশ্নের উত্তর দেখুন।

ooo **Problem-05** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 3rd (Marketing) 2018 No–6**

You are supplied with the following information :

Sampling Error (E) = 1.5, standard deviation (σ) = 4, critical value of z = 2.58, calculate the required sample size.

Solution : ৩৬৪ পৃষ্ঠার **Problem-05** নং প্রশ্নের উত্তর দেখুন।

ooo **Problem-06** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 3rd (Marketing) 2019 No–12**

A population has three elements that are 4, 6, 8, Select the number of sample size of 2 and show that sample mean is riskless determinant to population mean.

Solution : ৩৬৪ পৃষ্ঠার **Problem-06** নং প্রশ্নের উত্তর দেখুন।

ooo **Problem-07** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 2nd (Old) 2013 No–13; (Accounting) 2017 No–13; 3rd (Marketing) 2018 No–12**

Dhaka City Motors Ltd./Progoty Motors Ltd./Navana Motors Limited has 5 representative. The numbers of sold cars in the last month are given :

Representative	A	B	C	D	E
Production	5	15	10	25	20

(i) How many different samples are possible to take considering 2 at a time?
(ii) Record all possible samples and find its mean taking 2 at a time.
(iii) Compare between mean of sample means and population mean?

Solution : ৩৬৫ পৃষ্ঠার **Problem-07** নং প্রশ্নের উত্তর দেখুন।

[২০২৪ সালের পরীক্ষায় যা এসেছিল-

XX **Problem-08** ***Year Question & Solution*** ⊙ **N.U. BBA (Hons.) 3rd Marketing 2022 No–17; 2024 No–15**

Draw a sample of 3 units from a population of (4, 10, 3, 7, 8) and find :
(i) Write down the samples.
(ii) Obtain sample mean and population mean.
(iii) Show that mean of sample mean is equal to population mean.

Solution : ৩৬৬ পৃষ্ঠার **Problem-08** নং প্রশ্নের উত্তর দেখুন।

✪✪ **Problem-09** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 2nd (Finance) 2018 No–9**

Draw a sample of 3 units from a population of (2, 4, 6, 8). Write down samples. Show that variance mean of samples is equal to population variance.

Solution : ৩৬৭ পৃষ্ঠার **Problem-09** নং প্রশ্নের উত্তর দেখুন।

✪✪✪ **Problem-10** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 3rd (Marketing) 2019 No–11; 2021 No–12**

A population consists of the following values:

2, 2, 4, 4 and 8.

(i) List all samples of size 2 and compute the mean of each sample.

(ii) Compute the mean of the distribution of sample means and the population mean. Compare the two values.

(iii) Compare the dispersion in the population with that of the sample means.

Solution : ৩৬৯ পৃষ্ঠার **Problem-10** নং প্রশ্নের উত্তর দেখুন।

Chapter 04 Estimation of Parameters

(পরামিতি অনুমান)

Part-A : Brief Questions :

✪✪✪ 01. **What is statistical estimation? 99%**

[N.U. (Hon's) 3rd 2022 (Marketing)]

Ans : Statistical estimation is the process of using a sample statistic to estimate a population parameter.

[রকেট স্পেশাল সাজেশন্স : উপরিউক্ত ৯৯% প্রশ্নটি ১০০% গুরুত্বসহকারে পড়তে হবে। –সম্পাদক]

✪✪ 02. **What is estimator?**

Ans : A statistic used to estimate a population is called an estimator.

✪✪ 03. **What is point estimation?**

Ans : In statistic, point estimation involves the use of sample data to calculate a single value (known as a statistic) Which is to serve as a "best guess" or "best estimate" of an unknown population parameter.

✪✪✪ 04. **What is rejection region? 99%**

[N.U. (Hon's) 3rd 2016 (Marketing)]

Ans : The rejection region is the portion of the sampling distribution of a test statistic that corresponds to values for which the null hypothesis is rejected.

✪✪ 05. **Give the format of ANOVA table in one-way classification.**

[D.U. (Affiliated) (Hon's) 3rd 2016 (Marketing)]

Ans : In this case, only one variety is experimented, so it is called one factor experiment. The null hypothesis is accepted because there is no difference between treatment.

✪✪✪ 06. **What do you meant by interval estimate?**

Ans : An estimate of a population parameter given by two numbers between which the parameter may be considered to lie is called an interval estimate of the parameter

✪✪ 07. **Write down the formula of related efficiency?**

Ans : Related efficiency $= \frac{\text{Variance } [\hat{\theta}_2]}{\text{Variance } [\hat{\theta}_1]}$

✪✪✪ 08. **What is method of maximum likelihood?**

Ans : A general method for determining good estimators is called the method of maximum likelihood.

✪✪✪ 09. **Write down the formula of sample proportion?**

Ans : Sample proportion : $P = \frac{x}{n}$

Part-B & C: Short & Broad Questions :

✪✪ 01. What do you mean by confident limits for population mean? *[D.U. (Affiliated) (Hon's) 3rd 2016 (Marketing)]*

[Ans : ব্যবসায় পরিসংখ্যান মার্কেটিং ইন্টারেস্ট (২য় খণ্ড) বইয়ের 371 Page 4.01 No.]

Or, What are confidence limits? How are they determines?

✪✪✪ 02. (a) Explain clearly the desirable properties of a point estimate. **[Ans : 371 Page 4.03 No.]**

[Text Exam-Dhaka commerce, MM, Mirpur and Azam Khan Collage]

(b) What information and assumptions must be given to compute the sample size for an interval estimate of the universe mean? **[Ans : 372 Page 4.04 No.]**

০০০ 03. **What are the merits and limitation of multiple correlating analysis? ৯৯% [Ans : 372 Page 4.05 No.]**

[Test Exam-Edward, Bhala, Carmichael, Rajshahi, BL College]

০০০ (04) **What do you understand by estimation? In what sense, do you consider estimation as a procedure of decision-making? ৯৯% [Ans : 371 Page 4.02 No.]**

[Test Exam-City, Tongi and Tolaram College; N.U. (Hon's) 3rd 2016 (Marketing)]

[রকেট স্পেশাল সাজেশন্স : উপরিউক্ত ৯৯% প্রশ্নটি ১০০% গুরুত্ব সহকারে পড়তে হবে। -সম্পাদক]

Help Line ***Dear Student's, In above Suggestions in Part-B and C- 04 No. Question are most important for the Examinee of 2025.*** ***..... Editor***

গাণিতিক সমস্যাবলি ও সমাধান (Mathematical Problems & Solution)

Easy Math Basic Problem : Category-B & C

০০ **Problem-01** ***Basic Question & Solution***

A machine is producing ball bearings with diameter of 0.5 inches. It is know that the standard deviation of the ball bearing is 0.005 inches. A sample of 100 ball bearing is selected and their average diameter is found to be 0.498 inches. Determine the 99 percent confidence interval.

Solution : ব্যাতক্লশ মার্কেটিং ইন্সপেক্স (২য় খণ্ড) বইয়ের ৩৭৩ পৃষ্ঠার **Problem-01** নং প্রশ্নের উত্তর দেখুন।

০০ **Problem-02** ***Basic Question & Solution***

A sample of 150 items from machine A had an average life of 1,400 hrs. A similar sample of 100 items from machine B had a mean life of 1,200 hrs. Past records indicate that the standard deviation of the items produced by machine A is hrs. and by machine B is 80 hours. Find 95 per cent confidence limits on the difference in the average lifetimes of the populations of the items produced by the two machines.

Solution : ৩৭৫ পৃষ্ঠার **Problem-07** নং প্রশ্নের উত্তর দেখুন।

০০ **Problem-03** ***Basic Question & Solution***

A firm wishes to estimate with a maximum allowable error of 0.05 and a 95 per cent level of confidence, the proportion of consumers who prefer its product. How large a sample will be required in order to make such an estimate if the preliminary sales reports indicate that 25 per cent of all consumers prefer the firm's product ?

Solution : ৩৭৫ পৃষ্ঠার **Problem-08** নং প্রশ্নের উত্তর দেখুন।

০০ **Problem-04** ***Basic Question & Solution***

A ball pen manufacturer makes a lot of 10,000 refills. The procedure desires some control over these lots so that no lot will contain an excess number of defective refills. He decides to take a random sample of 400 refills for inspection from a lot of 10,000 and finds 9 defectives. Obtain a 90% confidence interval for the number of defectives in the entire lot.

Solution : ৩৭৬ পৃষ্ঠার **Problem-10** নং প্রশ্নের উত্তর দেখুন।

০০ **Problem-05** ***Basic Question & Solution***

A statistics consultant with the association of personal director was asked to determine what proportion of electrical personnel who change jobs do so because they are bored with their work. A random sample of 400 electrical personnel who had recently changed jobs were enquired and 200 state that they changed jobs because of bored. The statisticians prepared a 95 percent confidence interval for the true proportion changing jobs because of boredom. What are the lower and upper limits of this interval?

Solution : ৩৭৭ পৃষ্ঠার **Problem-14** নং প্রশ্নের উত্তর দেখুন।

○○ **Problem-06** ***Basic Question & Solution***

A random sample or six casting drawn from a universe of 75 castings shows the followings weight for each. Compute an interval estimate for μ at 2% level of significance.

Costing No.	1	2	3	4	5	6
Weight (Kg)	82.9	83.5	84.1	83.6	82.5	84.4

Solution : ৩৭৮ পৃষ্ঠার **Problem-16** নং প্রশ্নের উত্তর দেখুন।

Chapter 05 Test of Hypothesis

(কল্পনা যাচাই)

Part-A : Brief Questions & Solutions:

○○ 01 **Define hypothesis?** 99%

[Test Exam-Eden College (Acc.); D.U. (Affiliated) (2nd) 2017 (Finance); D.U. (Affiliated) (Hon's) 3rd 2016 (Marketing); N.U. (Hon's) 3rd 2019, 2021, 2023 (Marketing)]

Or, What is hypothesis?

Ans : Hypothesis is a statement about a population parameter developed for the purpose of testing.

রিকেট স্পেশাল সাজেশন্স : উপরিউক্ত ৯৯% প্রশ্নটি ১০০% গুরুত্বসহকারে পড়তে হবে। -সম্পাদক]

○○○ 02. **What is statistical hypothesis?**

Ans : A statistical hypothesis is a statement about the probability distribution of a set random variables.

03. What is null hypothesis. [২০২৪ সালে যা এসেছিল]

[Test Exam-Dhaka City (Acc.), Eden College (Mgt); N.U.BBA 2011; N.U. (Hon's) 3rd 2017, 2020, 2022, 2024 (Marketing)]

***Ans :** The hypothesis about a population parameter we wish to test is called a null hypothesis.*

○○○ 04. **What is critical value.** 99%

[Test Exam-Eden College (Acc.) N.U. (Hon's) 3rd 2016, 2018, 2020, 2022 (Marketing)]

Ans : The value of the standard statistic (Z or t) beyond which we reject the null hypothesis.

05. What is t-distribution? [২০২৪ সালে যা এসেছিল] [N.U. 2024 (Marketing)]

***Ans :** T-distribution is a probability distribution of the standardized sample mean when the population variance is unknown and the sample size is small.*

○○○ 06. **What is power of the test?**

Ans : The probability of not committing a type II error is called the power of the test hypothesis. *[N.U.BBA (Finance)-2018]*

○○ 07. **Define one tailed test.**

[Test Exam-Habibullahbaher College (Acc.)]

Ans : One-tailed test is a hypothesis test in which there is only one rejection region.

○○○ 08. **What is forecasting?**

[N.U.BBA (2nd)-2015 (Marketing); D.U. (2nd)-2017 (Marketing)]

Ans.: Forecasting is the process of predicting future events or conditions based on the analysis of past and present data.

○○○ 09. **What is two-tailed test?** *[N.U.BBA (Acc)-2015]*

Or, Define two tailed test.

Ans : When the hypothesis about the population mean is rejected because the test statistic falls into either tail of the sampling distribution, it is called two-tailed test.

○○ 10. **What is right tailed test?**

Ans : A test of any statistical hypothesis where the alternative is right tailed Such as Ho : $\mu = \mu_o$ Vs H1 : $\mu > \mu_o$ is called a right tailed test.

11. What is contingency table? [২০২৪ সালে যা এসেছিল] [N.U. 2024 (Marketing

***Ans :** The cross classification can be conveniently displayed means of a table is called contingency table.*

○○○ 12. **What is meant by type I error?** 99%

[Test Exam-Eden College; D.U. 2nd-(Finance)-2017; N.U. (Hon's) 3rd 2018 (Marketing)]

Or, Type I Error.

Ans : Type 1 error is the rejecting a null hypothesis when it is true.

○○ 13. **What do you mean by test statistic?**

Ans : Any mathematical function of the elements belong to a sample is called as statistic of that sample.

14 What is test of hypothesis? 99%

[N.U. (Hon's) 3rd 2016, 2018, 2021 (Marketing)]

Ans : Test of hypothesis is a statistical procedure that is used to provide evidence in favour of same statement.

[রকেট স্পেশাল সাজেশন্স : উপরিউক্ত ৯৯% প্রশ্নটি ১০০% গুরুত্বসহকারে পড়তে হবে। -সম্পাদক]

15 What do you mean by type II error? 99%

[N.U. (Hon's) 3rd 2016, 2019, 2021 (Marketing)]

Or, Define Type II Error.

Or, What is type-II Error?

Ans : Type II error is the accepting a null hypothesis when it is false.

[রকেট স্পেশাল সাজেশন্স : উপরিউক্ত ৯৯% প্রশ্নটি ১০০% গুরুত্বসহকারে পড়তে হবে। -সম্পাদক]

16. Write the formula of Z test

Ans : (i) $Z = \frac{\bar{x} - \mu}{\frac{S}{\sqrt{n}}}$ [For sample and population mean]

(ii) $Z = \frac{x_1 - x_2}{S\sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}$ [For two Sample mean]

(iii) $Z = \frac{P_1 - P_2}{\sqrt{Pq\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}}$ [For Sample Population]

17. What is Alpha?

Ans : The probability of Type 1 error.

18. What is Beta?

Ans : Beta (β) The probability of a Type II error.

19. What is raw scale?

Ans : Measurement in the variable's original units.

20. Who has developed t-test? 99% *[N.U. 2021]*

Ans. William Sealy Gosset developed t-test.

21 What is small sample test? 99%

[N.U. (Hon's) 3rd 2022 (Marketing)]

Ans : When sample size become at least 30 or not more then 30 then used method in statistical test of hypothesis is called small sample test.

[রকেট স্পেশাল সাজেশন্স : উপরিউক্ত ৯৯% প্রশ্নটি ১০০% গুরুত্বসহকারে পড়তে হবে। -সম্পাদক]

Part-B & C: Short & Broad Questions :

01. What is hypothesis testing? **[Ans : বাস্তবিক মার্কেটিং ইন্সপেক্স (২য় খণ্ড) বইয়ের 381 Page 5.01 No.]**

[N.U.BBA (old) 2006, 2007, 2009, 2010; D.U. (Affiliated) 2017 (Marketing-2nd)]

02. Distinguish between statistic and parameter. **[Ans : 387 Page 5.30 No.]** *[N.U.BBA (old) 2009]*

03. **Briefly describe the steps of testing hypothesis. 99% [Ans : 381 Page 5.06 No.]**

[Test Exam : Sathkhira College; N.U.BBA (old) 2011; (Hon's) 3rd 2016, 2019 (Marketing); 2017 (Accounting-2nd)]

Or, Discuss the procedure of testing a hypothesis.

04 What are the differences between null and alternative hypothesis? 99% [Ans : 386 Page 5.23 No.]

Or, Distinguish between null hypothesis and alternative hypothesis. *[N.U. 2017, 2019 (Marketing) 3rd]*

[রকেট স্পেশাল সাজেশন্স : উপরিউক্ত ৯৯% প্রশ্নটি ১০০% গুরুত্ব সহকারে পড়তে হবে। -সম্পাদক]

05. Describe the various steps involved in testing hypothesis. **[Ans : 387 Page 5.29 No.]**

[N.U. BBA Hon's-2nd (Finance)-2015, 2018; (Accounting-2nd)-2015; D.U. (Affiliated) (Hon's) 3rd 2016 (Marketing)]

Or, What steps are the followed in hypothesis testing?

06. Write short note : **[Ans : 382 Page 5,07 No.]**

(a) Hypothesis and hypothesis testing. *[N.U.BBA 2nd 2018(Acc)]*

(b) Two way classification of analysis of variance.

07. Where do you use students distributions? **[Ans : 385 Page 5.19 No.]**

08 Distinguish between type-I and type-II error. 99% [Ans : Page No-386 Ques. No-5.24]

[N.U. (Hon's) 3rd 2018, 2021, 2023 (Marketing)]

[রকেট স্পেশাল সাজেশন্স : উপরিউক্ত ৯৯% প্রশ্নটি ১০০% গুরুত্ব সহকারে পড়তে হবে। -সম্পাদক]

[২০২৪ সালের পরীক্ষায় যা এসেছিল–

9. Distinguish between z-test and t-test. **[Ans : Page No. 385; Ques. No. 5.20]**

[N.U. (Hon's) 3rd 2016, 2018, 2020, 2022, 2024 (Marketing); 2017 (Finance-2nd)]

Help Line ***Dear Student's, In above Suggestions in Part-B and C- 04 & 08 No. Question are most important for the Examinee of 2025.*** ***..... Editor***

গাণিতিক সমস্যাবলি ও সমাধান (Mathematical Problems & Solution)

Easy Math Basic Problem : Category-B & C

oo **Problem-01** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 2nd (Finance) 2022 No–17(b)**

A telecom service provider claims that individual customers pay an average Tk. 400 per month with standard deviation of Tk. 25. A random sample of 50 customers bills during a given month is taken with a mean of 250 standard deviation of 15. What to say with respect to the claim made by the service provider?

Solution : ঝটাফটকম মার্কেটিং ইঞ্জেকশন (২য় খণ্ড) বইয়ের ৩৯২ পৃষ্ঠার **Problem-04** নং প্রশ্নের উত্তর দেখুন।

oo **Problem-02** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 3rd (Marketing) 2023 No– 14**

A random sample of 100 students gave a mean weight of 58 kg with a standard deviation of 4 kg. Test the hypothesis that the mean weight in the population is 60 kg.

Solution : ৩৯১ পৃষ্ঠার **Problem-02** নং প্রশ্নের উত্তর দেখুন।

[রকেট স্পেশাল সাজেশন্স : উপরিউক্ত প্রশ্নটি ১০০% গুরুত্ব সহকারে পড়তে হবে। -সম্পাদক]

ooo **Problem-03** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 2nd (Old) 2012 No–09; D.U. 3rd (Mgt) 2016 No–7; N.U. BBA (Hon's) 3rd (Marketing) 2018 No–13; 2020 No–13**

Intelligence test on groups of boys and girls gives the following result :

	Mean	S.D	N
Girls	75	15	150
Boys	70	20	250

Is there a significant difference in the mean scores obtained by boys and girls?

Solution : ৩৯৩ পৃষ্ঠার **Problem-07** নং প্রশ্নের উত্তর দেখুন।

ooo **Problem-04** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 3rd (Marketing) 2023 No– 13**

Intelligence test on two groups of boys and girls gives the following result :-

	Mean	Standard deviation	Numbers
Girls	90	10	50
Boys	84	12	70

Determine 95% confidence interval of the difference in the mean scores of boys and girls?

Solution : ৩৯৪ পৃষ্ঠার **Problem-08** নং প্রশ্নের উত্তর দেখুন।

[রকেট স্পেশাল সাজেশন্স : উপরিউক্ত প্রশ্নটি ১০০% গুরুত্ব সহকারে পড়তে হবে। -সম্পাদক]

ooo **Problem-05** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 3rd (Marketing) 2020 No–16**

Random samples drawn from two places gave the following data relating to the heights of adults males :

	Barishal.	Khulna
No. of adults male in sample	1200	1500
Average height (in inches)	68.50	68.58
Standard deviation (inches)	2.5	3.0

Test at 5% level of significance, that the mean height is the same for adults males in two places.

Solution : ৩৯৫ পৃষ্ঠার **Problem-10** নং প্রশ্নের উত্তর দেখুন।

ooo **Problem-06** ***Year Question & Solution*** ⊙ **N.U. BBA (Hons.) 3rd Marketing 2022 No–14**

You are working as a purchase manager for a company. The following information has been supplied to you by two manufacturers of electric bulbs.

	Company-A	Company-B
Mean life	1300 hours	1288 hours
Standard deviation	82 hours	93 hours
Sample size	80	80

Which brand of bulbs are you going to purchase if you desire to take a risk of 5%?

Solution : ৩৯৬ পৃষ্ঠার **Problem-11** নং প্রশ্নের উত্তর দেখুন।

oo **Problem-07** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 2nd (Finance) 2016 No–15(b)**

The following data are obtained from and investigation :

	Sample-1	Sample-2
No. of cases	400	900
Mean wages	Tk. 47.40	Tk. 50.30
S.D. of the wages	Tk. 3.10	Tk. 3.30

Solution : ৩৯৬ পৃষ্ঠার **Problem-12** নং প্রশ্নের উত্তর দেখুন।

ooo **Problem-08** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 3rd (Marketing) 2019 No–14**

Two types of new cars produced by a company are tested for petrol mileage. One group consisting of 36 cars averaged 14 kms per litre, while the other consisting of 72 cars averaged 12 kms per litre.

(i) What test statistic is appropriate if :

$S_1^2 = 1.5$ and $S_2^2 = 2.0$?

(ii) Test whether there exists a significant difference in the petrol consumption of these two types of cars (use a ==0.01).

Solution : ৩৯৭ পৃষ্ঠার **Problem-13** নং প্রশ্নের উত্তর দেখুন।

ooo **Problem-09** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 3rd (Marketing) 2017 No–14**

A samble of 56 antique dealers in Bangladesh revealed the following sales last year :

Sales (in thousand Tk.) :	54–56	56–58	58–60	60–62	62–64	64–66
No. of firms	10	12	20	8	4	2

It is likely that the sample come from the population with an average yearly sales of Tk. 55,000.

Solution : ৩৯৮ পৃষ্ঠার **Problem-14** নং প্রশ্নের উত্তর দেখুন।

oo **Problem-10** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 2nd (Accounting) 2016 No–17**

A random sample 234 students from 450 students in Dhaka University and 171 students from 380 students in Rajshai University gave their opinion against smoking. Is the difference of opinion of two University students are significant at 5% significance level?

Solution : ৩৯৯ পৃষ্ঠার **Problem-16** নং প্রশ্নের উত্তর দেখুন।

ooo **Problem-11** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 3rd (Marketing) 2016 No–15**

Intelligence test on two groups of boys and girls gave the following results : (When the population variances is unknown and unequal)

	Mean	Standard deviation	Number
Boys	50	10	10
Girls	55	8	12

Is there significant difference in the mean scores of boys and girls?

Solution : ৪০৪ পৃষ্ঠার **Problem-18** নং প্রশ্নের উত্তর দেখুন।

[২০২৪ সালের পরীক্ষায় যা এসেছিল-

☒☒ **Problem-12** ***Year Question & Solution*** ⊙ **N.U. BBA (Hons.) 3rd Marketing 2021 No–14; 2024 No–16**

Mean and standard deviation of production of the two worker groups are given below :

	1st group	2nd group
Number of workers	34	20
Mean of production	50 tons	55 tons
Standard deviation	6 tons	5 tons

Is there any significant difference between the groups at 5% level of significance?

Solution : ৪০৫ পৃষ্ঠার **Problem-19** নং প্রশ্নের উত্তর দেখুন।

ooo **Problem-13** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 2nd (Accounting) 2015 No–16(b); 2017 No–17; (Finance) 2020 No–16(b); 3rd (Marketing) 2017 No–15**

Prices of shares (in taka) of a company on the different days in a month were found to be :
66, 65, 69, 70, 69, 71, 70, 63, 64, 68.
Test whether the mean price of the shares in the month is Tk. 65.
(5% level d.f 0.9 is 2.262)

Solution : ৪০৭ পৃষ্ঠার **Problem-22** নং প্রশ্নের উত্তর দেখুন।

[২০২৪ সালের পরীক্ষায় যা এসেছিল-

☒☒ **Problem-14** ***Year Question & Solution*** ⊙ **N.U. BBA (Hons.) 3rd Marketing 2024 No–12**

In a recent interview on banking profession, 32 out of 150 male banking professionals rated the banking career as 'Excellent' and 17 out of 80 female banking professional rated it as 'Excellent'. Using 10% level of significance, can we conclude that there is a significant difference in the proportion of male versus female banking professionals who rate the banking career as excellent?

Solution : ৫৪৫ পৃষ্ঠার **Question No-12** নং প্রশ্নের উত্তর দেখুন।

মাস্টার্স (অনার্স-৩য় বর্ষ)-২২(খ)

Chapter 06 Small Sampling Theory

(ক্ষুদ্র নমুনায়ন তত্ত্ব)

Part-A : Brief Questions:

ooo 01. **What is experiment?** 99%

Ans : The activity that results in or produces an event is called experiment.

ooo 02. **What is meant by small sample?** 99%

Ans : If the sample units in a population are less than 30 or $n < 30$ then the sample is said to be small sample.

ooo (03) **What is sampling?** 99%

Ans : The methods for selecting sample from a population is known as sampling.

[রকেট স্পেশাল সাজেশন্স : উপরিউক্ত ৯৯% প্রশ্নটি ১০০% গুরুত্বসহকারে পড়তে হবে। -সম্পাদক]

ooo 04. **Write down the another name of 't' test?**

Ans : Student's test.

oo 05. **What is raw scale?**

Ans : It is the measurement in the variables original units.

ooo 06. **What does $\bar{d}$ mean?**

Ans : $\bar{d}$ means the mean difference.

oo 07. **Write the elaborating of D. F.**

Ans : Degree of freedom.

ooo 08. **What is the condition for small sample?**

Ans : No. Sample N < 30.

Part- B & C : Short & Broad Questions :

oo 01. Discuss the causes of non-sampling error? *[Test Exam-Edward, Chuadanga, MM and Satkhira College]*

[Ans মার্কেটিং সাজেশন্স (২য় খণ্ড) বইয়ের **410 Page 6.02 No.]**

ooo 02. **Describe the Difference/ Distinction/ between simple and weighted average.** 99%

(সাধারণ গড় ও ভারযুক্ত গড়ের পার্থক্য বর্ণনা কর।)**[Ans : 410 Page 6.03 No.]**

ooo 03. What is measures of dispersion? (বিস্তার পরিমাপ কি?) **[Ans : 411 Page 6.05 No.]**

Or, What do you mean by measures of dispersion? বিস্তার পরিমাপ বলতে কি বুঝ?

oo 04. Explain systematic random sampling. **[Ans : 410 Page 6.04 No.]**

[Test Exam : Kobi Nzarul, Sathkhira College; N.U.BBA (old) 2011]

Help Line ***Dear Student's, In above Suggestions in Part-B and C- 02 No. Question are most important for the Examinee of 2025.*** ***..... Editor***

গাণিতিক সমস্যাবলি ও সমাধান (Mathematical Problems & Solution)

Easy Math Basic Problem : Category-B & C

oo **Problem-01** ***Basic Question & Solution***

A random sample of 18 workers gave a mean weight of 48 kg with a standard deviation of 5 kg. Test the Hypothesis that the mean weight in the population is 50 kg.

Solution : মার্কেটিং সাজেশন্স (২য় খণ্ড) বইয়ের ৪১৩ পৃষ্ঠার **Problem-02** নং প্রশ্নের উত্তর দেখুন।

oo **Problem-02** ***Basic Question & Solution***

The standard deviation of the weight of 100 gm bread made by a certain bakery is 1gm. On a certain day the owner doubted that the production is out f control. To check whether its production is under control, employees select a random sample of 25 breads and find that their mean weight is 99.5 gm. Test the doubt of the owner at 5% level of significance.

Solution : ৪১৩ পৃষ্ঠার **Problem-03** নং প্রশ্নের উত্তর দেখুন।

ooo **Problem-03** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 3rd (Marketing) 2020 No–17**

The Drug International Company wishes to test whether its three salesmen X, Y and Z tend to make sales of the same size or whether they differ in their selling ability as measured by the average size of their sales. During the last week, out of 14 sales, X made 5, Y made 4 and Z made 5 calls. The following are the weekly sales (in Tk. thousand) record of three salesmen :

X	Y	Z
300	600	700
400	300	300
300	300	400
500	400	600
0	-	500

Test whether the three salesmen's average sales differ in size.

Solution : ৪১৪ পৃষ্ঠার **Problem-04** নং প্রশ্নের উত্তর দেখুন।

[রকেট স্পেশাল সাজেশন্স : উপরিউক্ত প্রশ্নটি ১০০% গুরুত্ব সহকারে পড়তে হবে। -সম্পাদক]

oo **Problem-04** ***Year Question & Solution*** ⊙ **D.U. BBA (Hon's) 3rd (Marketing) 2016 No–16**

A company is considering two different television advertisements for promotion of a new product. Management believes that advertisement A is more effective than advertisement B. Two test market areas with virtually identical consumer characteristics are selected: advertisement A is used in one area and advertisement B in the other area. In a random sample of 60 customers who saw advertisement A, 18 ried the product. In a random sample of 100 customers who saw advertisement B, 22 tried the product. Does this indicate that advertisement A is more effective than advertisement B, if a 5% level of significance is used?

Solution : ৪১৬ পৃষ্ঠার **Problem-07** নং প্রশ্নের উত্তর দেখুন।

oo **Problem-05** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 3rd (Marketing) 2017 No–13**

In atest given to two groups of students, the marks obtianed are as follows :

First Group :	18	20	35	50	45	35	34	46	42
Second Group :	25	28	26	35	34	45	47	40	38

Examine the significance of difference between the arithmetic mean of the marks secured by the students of the above two groups. The table value : at 5% level of significance is 2.14.

Solution : ৪১৭ পৃষ্ঠার **Problem-08** নং প্রশ্নের উত্তর দেখুন।

Chapter 07 Chi-Spuare Test

(কাই-বর্গ যাচাই)

Part-A : Brief Questions :

01 **What is chi-square test?** 99%

[N.U. (Hon's) 3rd 2016, 2018 (Marketing)]

Or, What do you mean by χ^2 (Chi-square)?

Ans : χ^2 (Chi-square) is a statistical method of justifying whether the difference between the expected value from a binominal or multistage distribution and the observed value from the distribution is meaningful or not.

[রকেট স্পেশাল সাজেশন্স : উপরিউক্ত ৯৯% প্রশ্নটি ১০০% গুরুত্বসহকারে পড়তে হবে। -সম্পাদক]

ooo 02. **What is considered standard significance level?**

Ans : Generally comparison is made at 5% level of significance. Because 5% is considered as standard significance level.

ooo 03. **When null hypothesis is rejected?**

Ans : If the calculated value is equal or greater than the theoretical value, it means the different between the expected and observed value is significant. In this case the null hypothesis is rejected.

04 **What is permutation?** 99%

[N.U. (Hon's) 3rd 2016, 2020 (Marketing)]

Ans : A permutation is an arrangement or ordering of a number of distinct objects.

[রকেট স্পেশাল সাজেশন্স : উপরিউক্ত ৯৯% প্রশ্নটি ১০০% গুরুত্বসহকারে পড়তে হবে। -সম্পাদক]

ooo 05. **Write the formula of chi-square test?** 99%

[N.U. (Hon's) 2022; D.U. (Affiliated) 2017 (Accounting)]

Or, Write down the formula of χ^2 test.

Ans : $\chi^2 = \sum \frac{(O-E)^2}{E}$

06. What do you mean by chi-square distribution? [২০২৪ সালে যা এসেছিল]

[N.U. (Hon's) 3rd 2016, 2019, 2021, 2024 (Marketing)]

Or, What is chi-square distribution?

Or, Define the chi-square (χ^2) distribution.

Ans : The distribution of the chi-square statistic is called the chi-square (χ^2) distribution.

ooo 07. **Who used chi-square (χ^2) test first?**

Ans : Karl Pearson first use χ^2 test.

oo 08. **When Karl Pearson developed (χ^2) (chi-square) test?**

Ans : Karl Pearson developed this test in 1990.

oooo (09) **What is multiple correlation?** 99%

[N.U. (Hon's) 3rd 2019 (Marketing)]

Ans. If we study the relationship among three or more variables, then it is called multiple correlation.

[রকেট স্পেশাল সাজেশন্স : উপরিউক্ত ৯৯% প্রশ্নটি ১০০% গুরুত্বসহকারে পড়তে হবে। -সম্পাদক]

ooo 10. **What are the uses of χ^2 -test?**

Ans : The uses of x^2 are :

(i) To test the normal Population,

(ii) To test the goodness of fit,

(iii) To test homogeneity between expected and observe f frequency.

ooo 11. **What does $\propto$ mean?**

Ans : The symbol $\propto$ means the probability of a type-I error.

oo 12. **What does β mean?**

Ans : The symbol β means the probability of type-II error.

ooo 13. **Write the formula of Co-efficient of Contingency.**

Ans : $C = \sqrt{\frac{x^2}{x^2+n}}$

Part- B & C : Short & Broad Questions :

ooo 01. What is χ^2 test? [Ans মার্কেটিং হাইলাইটস (২য় খণ্ড) বইয়ের 421 Page 7.01 No.]

oo 02. Show the difference between diagram and graph. **[Ans : 422 Page 7.08 No.]**

[Test Exam-Chuadanga, Bhola, City, Rangpur, Mirpur and Azizul College; N.U.BBA - 2nd 2015]

ooo (03) **Discuss the properties of chi-square test.** 99% **[Ans : 422 Page 7.05 No.]**

[Text Exam-Eden, City, Rajbair and MM Collage; N.U. (Hon's) 3rd 2017, 2021 (Marketing)]

Or, What are the uses of chi-square test?

[রকেট স্পেশাল সাজেশন্স : উপরিউক্ত ৯৯% প্রশ্নটি ১০০% গুরুত্ব সহকারে পড়তে হবে। -সম্পাদক]

ooo 04. Explain the conditions for the application of $\chi 2$ (Chi-Square) test. **[Ans : 421 Page 7.03 No.]**

ooo 05. Write is the difference or distinction between parametic test and non-parametic test. **[Ans : 421 Page 7.04 No.]**

ooo 06. **Discuss the uses of $\chi 2$ -test or application of $\chi 2$ -test or Chi-square test?** 99% **[Ans : 422 Page 7.06 No.]**

Help Line ***Dear Student's, In above Suggestions in Part-B and C- 03 & 06 No. Question are most important for the Examinee of 2025.*** ***..... Editor***

গাণিতিক সমস্যাবলি ও সমাধান (Mathematical Problems & Solution)

Easy Math Basic Problem : Category-B & C

ooo **Problem-01** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 3rd (Marketing) 2019 No–07; 2021 No–08**

800 people are involved in a factory. Out of them 350 are educated. In an efficiency test, 175 are terminated for their inefficiency and 45 of them are educated. Make a contingency table using the above data.

Solution : মার্কেটিং হাইলাইটস (২য় খণ্ড) বইয়ের ৪২৫ পৃষ্ঠার **Problem-02** নং প্রশ্নের উত্তর দেখুন।

০০০ Problem-02 Year Question & Solution ⊙ N.U. BBA (Hons.) 3rd Marketing 2022 No–16

From the following data, test the significance of relation between facebook user and age at 5% level of significance :

	Young	Old
Facebook user	18	28
Facebook nonuser	15	35

Solution : ৪২৬ পৃষ্ঠার **Problem-04** নং প্রশ্নের উত্তর দেখুন।

০০০ (৯৯%) [রকেট স্পেশাল সাজেশন্স : উপরিউক্ত প্রশ্নটি ১০০% গুরুত্ব সহকারে পড়তে হবে। –সম্পাদক]

০০০ Problem-03 Year Question & Solution ⊙ N.U. BBA (Hon's) 3rd (Marketing) 2020 No–14

From the following data test the significance of relation between smoking and drinking :

	Drinking	Not-drinking
Smoking	18	26
Not-smoking	10	30

Solution : ৪২৮ পৃষ্ঠার **Problem-06** নং প্রশ্নের উত্তর দেখুন।

০০০ Problem-04 Year Question & Solution ⊙ N.U. BBA (Hon's) 3rd (Marketing) 2018 No–15

From the following data, test the significance of relation between education and skillness :-

	Skilled	Unskilled
Educated	80	15
Uneducated	20	35

Solution : ৪২৯ পৃষ্ঠার **Problem-07** নং প্রশ্নের উত্তর দেখুন।

০০০ Problem-05 Year Question & Solution ⊙ N.U. BBA (Hon's) 3rd (Marketing) 2016 No–13; 2019 No–15; 2021 No–16; 2023 No– 15

From the following data, test the significance of relation between education and skillness at 5% level of significance :

	Skilled	Un-skilled
Educated	75	20
Un-educated	25	30

Solution : ৪২৯ পৃষ্ঠার **Problem-08** নং প্রশ্নের উত্তর দেখুন।

০০০ Problem-06 Year Question & Solution ⊙ N.U. BBA (Hons.) 3rd Marketing 2022 No–15

The following table gives the numbers of good and bad parts produced by each of three shifts in a factory.

Shift	Good	Bad
Day	900	130
Evening	700	170
Night	400	200

If there any association between the shift and equality of parts produced?

Solution : ৪৩০ পৃষ্ঠার **Problem-09** নং প্রশ্নের উত্তর দেখুন।

○○○ **Problem-07** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 3rd (Marketing) 2016 No–16**

The following figures show the number of sales per day in a showroom of Sony company :

Day-	1	2	3	4	5	6	7	8	9	10	Total
Frequency	12	8	20	2	14	10	15	6	9	4	100

Test whether these frequencies in agreement with the belief that sales conditions were the same during this 10-day period.

Solution : ৪৩১ পৃষ্ঠার **Problem-10** নং প্রশ্নের উত্তর দেখুন।

○○ **Problem-08** ***Basic Question & Solution***

From the adult male population of seven large cities random sample of married and unmarried men as given below were taken. Can it be said there is a significant variation among the people of different cities in the tendency to marry?

City	A	B	C	D	E	F	G	Total
Married	170	285	165	106	153	125	146	1,150
Unmarried	40	125	35	37	55	35	33	360
Total	210	410	200	143	208	160	179	1,510

Given for $v = 6$, $\chi^2\ 0.05 = 12.6$

Solution : ৪৩২ পৃষ্ঠার **Problem-11** নং প্রশ্নের উত্তর দেখুন।

○○○ **Problem-09** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 3rd (Marketing) 2020 No–09**

Heights in inches of 10 students are given below :
38, 40, 45, 53, 47, 43, 55, 48, 52, 49.
Test the variance of the distribution of height of all students from which the above sample of 10 students was drawn is equal to 20 square inches.

Solution : ৪৩৪ পৃষ্ঠার **Problem-13** নং প্রশ্নের উত্তর দেখুন।

○○ **Problem-10** ***Basic Question & Solution***

Of the 1,000 workers in a factory exposed to an epidemic, 700 in all were attacked, 400 had been inoculated and of these, 200 were attacked. On the basis of this information, can it be said that inoculation and attack are independent?

Solution : ৪৩৬ পৃষ্ঠার **Problem-16** নং প্রশ্নের উত্তর দেখুন।

○○ **Problem-11** ***Basic Question & Solution***

The following figures show the distribution of digits in numbers chosen at random from a telephone director) :

Digit :	0	1	2	3	4	5	6	7	8	9	Total
Frequency :	1026	1107	997	966	1075	933	1107	972	964	853	10,000

Test, whether the digits may be taken to occur equally frequently in the directory.

Solution : ৪৩৮ পৃষ্ঠার **Problem-18** নং প্রশ্নের উত্তর দেখুন।

○○ **Problem-12** ***Basic Question & Solution***

Two sample polls of votes for two candidates A and B for a public office are taken, one from among residents of rural area and one from urban areas. The results are given below. Examine, whether the nature of the area is related o the voting preference in this election.

Votes for / Area	A	B	Total
Rural	620	380	1000
Urban	550	450	1000
Total	1170	830	2000

Solution : ৪৪০ পৃষ্ঠার **Problem-22** নং প্রশ্নের উত্তর দেখুন।

✪✪✪ **Problem-13** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 3rd (Marketing) 2017 No–16**

Of the 500 workers in a factory exposed to an epidemic 350 in all were attacked, 200 had been inoculated and of these 100 were attacked. Set out the data in a tabular form and find out whether in the given case the addociation is or is not.significant.

Solution : ৪৪২ পৃষ্ঠার **Problem-24** নং প্রশ্নের উত্তর দেখুন।

Chapter 08 Analysis of Variance

(ভেদাংঙ্ক বিশ্লেষণ)

Part-A : Brief Questions & Solutions :

✪✪✪ (01) **What is analysis of variance?** 99%

[N.U. (Hon's) 3rd 2017, 2021, 2023 (Marketing); D.U. (Affiliated) (Hon's) 3rd 2016, 2019, 2021 (Marketing)]

Ans : The analysis of variance is a technique that test for significant of the different among more than two sample means.

[রকেট স্পেশাল সাজেশন্স : উপরিউক্ত ৯৯% প্রশ্নটি ১০০% গুরুত্ব সহকারে পড়তে হবে। -সম্পাদক]

✪✪✪ 02. **What is meant by one way classification of data?** 99%

Ans : If the data are classified according to only one criterion or factor then the classification is called one way classification.

✪✪ 03. **Give two applications of analysis of variance.**

Ans : (i) Every type of experimental design. (ii) Agricultural blocks and treatments.

✪✪✪ 04. **What do you understand by one way analysis of variance?**

Ans : The term one factor analysis of variance refers to the fact that a single variance or factor of interest s controlled and its effect on the elementary units of observed.

✪✪ 05. **What is two way classification in analysis of variance?**

Ans : Two way classification in analysis of variance is a technique that can test two sets of hypothesis with same data at the same time.

✪✪✪ 06. **Who fast introduce the concept of analysis of variance?**

Ans : The term 'Analysis of Variance' was introduced by Prof. R.A. Fisher.

✪✪✪ 07. **What are the sources of variation?**

Ans : The total variation in any set of data is due to a number of causes which may be classified as : i. Variation due to Assignable, ii. Variation due to change causes

✪✪ 08. **What are variation of assignable causes?**

Ans : The variation which may be occurred by the causes of factors (treatments) or known causes is call variation due to assignable causes.

✪✪✪ 09. **What are variation of change causes?**

Ans : The inherent variation of the random variable within the observations of a class is known as variation due to chance causes.

✪ 10. **Who developed the technique of analysis of variance?**

Ans : R.A Fisher.

11. What is the full meaning of ANOVA? [২০২৪ সালে যা এসেছিল] [N.U. (Hon's) 2022, 2024]

Ans : Analysis of Variance.

Part-B & C : Short & Broad Questions :

✪✪✪ 01. (a) **What do you mean by analysis of variances?** 99%

[Ans মার্কেটিং ইনিশিয়াস (২য় খণ্ড) বইয়ের 448 Page 8.01 No.]

(b) Discuss the various assumptions in analysis of variance. 99% **[Ans : 448 Page 8.02 No.]**

Or, What are the assumption to analysis of variance? *[D.U. (Affiliated) (Hon's) 3rd 2016 (Marketing)]*

✪✪ 02. What do mean by degree of freedom? Explain it with examples. **[Ans : 449 Page 8.05 No.]**

✪✪ 03. (a) What are the major assumption of ANOVA? **[Ans : 449 Page 8.10 No.]** *[N.U. BBA Hon's-2nd (Finance)-2018]*

(b) Describe the steps followed in analysis of variance. **[Ans : 448 Page 8.04 No.]**

[N.U. BBA Hon's-2nd (Finance)-2018; D.U. (Affiliated) (Hon's) 3rd 2016 (Marketing)]

Or, Briefly explain the procedure followed in analysis of variance.

Help Line ***Dear Student's, In above Suggestions In Part-B and C- 01 No. Question are most Important for the Examinee of 2025.*** ***..... Editor***

গাণিতিক সমস্যাবলি ও সমাধান (Mathematical Problems & Solution)

Easy Math Basic Problem : Category-B & C

০০ Problem-01 *Basic Question & Solution*

A random sample is selected from each of three makes of rope their breaking strength (in pounds) are measured, with the following results :

I	II	III
70	100	60
72	110	65
75	108	57
80	112	84
83	113	87
	120	73
	107	

Test, whether the breaking strength of the rope differ significantly.

Solution : মার্কেটিং এক্সপ্রেস (২য় খণ্ড) বইয়ের ৪৫০ পৃষ্ঠার **Problem-01** নং প্রশ্নের উত্তর দেখুন।

০০ Problem-02 *Basic Question & Solution*

Following table gives the number of refrigerators sold by 4 salesman in three months May, June and July.

	Salesman			
Month	A	B	C	D
May	50	40	48	29
June	46	48	50	45
July	39	44	40	39

Is there a significant difference in the sales made by the four salesmen?

Solution : ৪৫২ পৃষ্ঠার **Problem-03** নং প্রশ্নের উত্তর দেখুন।

[২০২৪ সালের পরীক্ষায় যা এসেছিল-

⊠⊠ Problem-03 *Year Question & Solution* ⊙ N.U. BBA (Hons.) 2nd (Acc.) 2015 No–17; 3rd (Marketing) 2017 No–17; 2024 No–17

Four salesmen were posted in different areas by a company. The number of units sold by them is given :

A	20	23	28	29
B	25	32	30	21
C	23	28	35	18
D	15	21	19	25

From the above information, can it be concluded that there is a significant difference in the performance of salesmen at 5% level of significance?

Solution : ৪৫৫ পৃষ্ঠার **Problem-06** নং প্রশ্নের উত্তর দেখুন।

০০ **Problem-04** ***Basic Question & Solution***

The Amit Merchandise Company wishes to test whether its three salesman A, B and C tend to make sales of the same size or whether they differ in their selling ability as measured by the average size of their sales. During the last week, out of 14 sales, A made 5, B made 4 and C made 5 calls. The following are the weekly sales (in Tk. thousand) record of three salesman :

A	B	C
300	600	700
400	300	300
300	300	400
500	400	600
0	---	500

Test, whether the three salesman's average sales differ in size.

Solution : ৪৫৮ পৃষ্ঠার **Problem-08** নং প্রশ্নের উত্তর দেখুন।

০০০ **Problem-05** ***Year Question & Solution*** ⊙ **D.U. BBA (Hon's) 2nd (Finance) 2017 No–17**

General Motors is a local company manufacturing Automobile Parts with a view to grabbing the domestic market of Bangladesh. Following table reveals the number of Automobile parts sold by four salesmen of General Motors in three consecutive seasons during this year :

Seasons	Salesmen				
	X	Y	Z	P	Total
Summer	36	36	21	35	128
Winter	28	29	31	32	120
Monsoon	26	28	29	29	112
Total	90	93	81	96	360

On the basis of this information, thirty out an analysis of variance using ANOVA table to determine whether there is any difference in the average sales made by four salesmen.

Solution : ৪৬১ পৃষ্ঠার **Problem-11** নং প্রশ্নের উত্তর দেখুন।

[২০২৪ সালের পরীক্ষায় যা এসেছিল-

☒☒ **Problem-06** ***Year Question & Solution*** ⊙ **N.U. BBA (Hons.) 3rd Marketing 2024 No–13**

To study the performance of three detergents and three different temperatures, the following 'Whiteness' readings were obtained with specially designed equipment :

Water temperature	Detergent		
	A	B	C
Cold	57	55	67
Warm	49	52	68
Hot	54	46	58

Using 5% level of significances, perform a two-way analysis of variance.

Solution : ৫৪৭ পৃষ্ঠার **Question No-13** নং প্রশ্নের উত্তর দেখুন।

Chapter 09 Partial and Multiple Correlation and Regression

(আংশিক ও বহুগুণক সহ-সম্বন্ধ নির্ভরণ)

Part-A : Brief Questions :

01 **What is partial correlation?** 99%

[N.U. (Hon's) 3rd 2018, 2021, 2023 (Marketing)]

Ans : Partial correlation is measures the strength and direction of a linear relationship between two continuous variables while controlling for the effect of one or more other continuous variables (also known as covariates or control variables).

[রকেট স্পেশাল সাজেশন্স : উপরিউক্ত ৯৯% প্রশ্নটি ১০০% গুরুত্বসহকারে পড়তে হবে। -সম্পাদক]

02 **What is coefficient of correlation?** 99%

[N. U. BBA (Hon's) (Marketing)(3rd)-2019]

Or, Define co-efficient of correlation.

Ans : Co-efficient of correlation is the measure of the strength of the linear relationship between two variables.

[রকেট স্পেশাল সাজেশন্স : উপরিউক্ত ৯৯% প্রশ্নটি ১০০% গুরুত্বসহকারে পড়তে হবে। -সম্পাদক]

০০ 03. Why correlation is used?

Ans : "Correlation analysis attempts to determine the degree of relationship among variables".

০০০ 04. Define zero correlation?

Ans : When there is no correlation between two variables is called zero correlation.

০ 05. What are the different types of correlation?

Ans : Three types correlation.

(i) Positive correlation;

(ii) Negative correlation;

(iii) Zero correlation.

০০০ 06. What is regression?

Ans : The general process of predicting one variable from another by statistical means, using previous data.

07. What is multiple regression? [২০২৪ সালে যা এসেছিল]

[N. U. (Acc) 2015; 2020, 2022, 2024 (Marketing)]

Or, What is meant by multiple regression?

Ans : When there is more than one independent variables then it is called multiple regression.

০০০ 08. What is regression line?

Ans : A line fitted to a set of data points to estimate the relationship between two variables.

০০০ 09. What is scatter diagram? 99%

[Test Exam-Dhaka (Acc.), Dhaka Commerce, Habibullahbaher, Shaikh Burhanuddin College; N. U. BBA (Hon's) (All) 2014, (Acc) 2017; (3rd)-2018 (Management); D.U. (3rd)-2016 (Management)]

Ans : The simplest device for studying correlation in two variables is a special type of a chart called scatter diagram.

০০০ 10. What is regression model? *[N. U. (Finance)-2018]*

Ans : A regression model is used to investigate the relationship between two or more variable and estimate one variable based on the others.

০০০ 11. What is regressions analysis? 99%

[N. U. BBA (Hon's) (All Dept.) 2011]

Ans : Regression Analysis is a mathematical measure of the average relationship between two or more variables in terms of original units of the data.

12 **What is meant by r = 1, r = – 1 and r = 0?** 99%

[N. U. BBA (Hon's) 2nd (All) 2012, 2014, (Acc) 2017; (3rd)-2017 (Management); Acc 3rd-2019]

Ans : The value of r = 1, Means the perfect positive correlation.

The value of r = –1, Means the perfect negative correlation.

The value of r = 0 Means called Zero (0) correlation.

[রকেট স্পেশাল সাজেশন্স : উপরিউক্ত ৯৯% প্রশ্নটি ১০০% গুরুত্বসহকারে পড়তে হবে। -সম্পাদক]

০০০ 13. What is Slope?

Ans : A constant for any given straight line whose value represents how much each unit change of the independent variable changes the dependent variable.

Part- B & C : Short & Broad Questions :

০০ 01. Discuss the partial correlation? [Ans : বাংলাদেশ মার্কেটিং ইনস্টিটিউট (২য় খণ্ড) বইয়ের 464 Page 9.01 No.]

[Test Exam-Feni, Rajbari, BL and Noakhali College; N.U.(2nd) 2015]

০০ 02. Briefly discuss the partial correlation co-efficient? [Ans : 464 Page 9.02 No.]

০০০ 03. What are merits of multiple correlation analysis? [Ans : 464 Page 9.04 No.]

[Test Exam-MM, Edward and Tolaram College]

○○ 04. Write down the significance of measuring correlation. **[Ans : 466 Page 9.09 No.]**

[Test Exam- Bhola, Tolaram, Shah Sultan, Notor, Mirpur and BL College; N.U.BBA (old) 2014; 2nd 2015]

○○○ (05) **Discuss the difference between correlation and regression.** 99% **[Ans : 467 Page 9.14 No.]**

[Text Exam-Sirjgonj, Azizul and Commerce Collage; N.U. (Old) 2013; 2016 (Finance-2nd); 2016 (Markating-2nd)]

Or, Distinguish between correlation and regresslon.

[রকেট স্পেশাল সাজেশন্স : উপরিউক্ত ৯৯% প্রশ্নটি ১০০% গুরুত্ব সহকারে পড়তে হবে। -সম্পাদক]

○○ 06. What are the assumptions of linear multiple regression and analysis? **[Ans : 467 Page 9.12 No.]**

[Text Exam-Barisal, Khulna, Tolaram, Dhaka commerce Collage]

○○○ 07. Explain the concept of multiple regression and try to find out an example in practical field where multiple regression analysis is likely to be helpful. *[N.U. (Hon's) 3rd 2016 (Marketing)]*

○○○ (08) **What are the uses of partial correlation?** 99% **[Ans : 464 Page 9.03 No.]**

Or, What are the uses of partial correlation analysis? *[N.U. (Hon's) 3rd 2016, 2018 (Marketing)]*

[রকেট স্পেশাল সাজেশন্স : উপরিউক্ত ৯৯% প্রশ্নটি ১০০% গুরুত্ব সহকারে পড়তে হবে। -সম্পাদক]

Help Line ***Dear Student's, In above Suggestions in Part-B and C- 05 & 08 No. Question are most important for the Examinee of 2025.*** ***..... Editor***

গাণিতিক সমস্যাবলি ও সমাধান (Mathematical Problems & Solution)

Easy Math Basic Problem : Category-B & C

○○○ **Problem-01** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's), 3rd (Marketing) 2017 No–07; 2021 No–09**

The following zero-order, correlation coefficients are given :

$r_{12} = 0.98$, $r_{13} = 0.44$ and $r_{23} = 0.54$

Calculate multiple correlation coefficient treating first variable as dependent and second and third variables as independent.

Solution : মার্কেটিং ইনসাইড (২য় খণ্ড) বইয়ের ৪৬৮ পৃষ্ঠার **Problem-02** নং প্রশ্নের উত্তর দেখুন।

○○○ **Problem-02** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 3rd (Marketing) 2019 No–9**

Is the following information correct or consistent?

$r_{12} = 0.07$, $r_{13} = -0.6$, $r_{23} = 0.9$

Solution : ৪৬৯ পৃষ্ঠার **Problem-04** নং প্রশ্নের উত্তর দেখুন।

○○○ **Problem-03** ***Year Question & Solution*** ⊙ **N.U. BBA (Hons.) 3rd Marketing 2022 No–9**

If $r_{12} = 0.8$, $r_{13} = -0.4$ and $r_{23} = -0.56$.

Find the value of $r_{12,3}$.

Solution : ৪৬৯ পৃষ্ঠার **Problem-05** নং প্রশ্নের উত্তর দেখুন।

○○○ (৯৯%) [রকেট স্পেশাল সাজেশন্স : উপরিউক্ত প্রশ্নটি ১০০% গুরুত্ব সহকারে পড়তে হবে। -সম্পাদক]

[২০২৪ সালের পরীক্ষায় যা এসেছিল-

☒☒ **Problem-04** ***Year Question & Solution*** ⊙ **N.U. BBA (Hons.) 3rd Marketing 2024 No–5**

If $r_{12} = 0.86$; $r_{13} = 0.65$ and $r_{23} = 0.72$; find the partial correlation co-efficient of $r_{12.3}$.

Solution : ৫৪৪ পৃষ্ঠার **Question No-5** নং প্রশ্নের উত্তর দেখুন।

○○○ Problem-05 Year Question & Solution ⊙ N.U. BBA (Hon's) 3rd (Marketing) 2018 No–16

The table shows the corresponding values of three variable X_1, X_2 and X_3. Find the least square regression of X_3 on X_1 and X_2. Estimate X_3 when $X_1 = 10$ and $X_2 = 6$:

X_1	3	5	6	8	12	14
X_2	16	10	7	4	3	2
X_3	90	72	54	42	30	12

Solution : ৪৭২ পৃষ্ঠার **Problem-09** নং প্রশ্নের উত্তর দেখুন।

○○○ Problem-06 Year Question & Solution ⊙ N.U. BBA (Hon's) 3rd (Marketing) 2020 No–15; 2023 No– 16

By using the following information, find the regression equation of X_1 on X_2 and X_3 :

X_1	12	22	32	28
X_2	6	12	16	22
X_3	4	6	12	18

Solution : ৪৭৩ পৃষ্ঠার **Problem-10** নং প্রশ্নের উত্তর দেখুন।

○○○ (৯৯%) [রকেট স্পেশাল সাজেশন্স : উপরিউক্ত প্রশ্নটি ১০০% গুরুত্ব সহকারে পড়তে হবে। –সম্পাদক]

○○○ Problem-07 Year Question & Solution ⊙ N.U. BBA (Hon's) 3rd (Marketing) 2019 No–16

By using the following information :

Performance evaluation (X_1)	28	33	21	40	38	46
Aptitude Test Score (X_2)	74	87	69	69	81	97
Prior Experience (X_3)	5	11	4	9	7	10

(i) Develop the estimating equation best describing these data.

(ii) If an employee scored 84 on the aptitude test and had a prior experience of 7 years, what performance evaluation would be expected?

Solution : ৪৭৫ পৃষ্ঠার **Problem-11** নং প্রশ্নের উত্তর দেখুন।

○○ Problem-08 Basic Question & Solution

In a trivariate distribution :

$\bar{x}_1 = -28.02$, $\bar{x}_2 = 4.91$, $\bar{x}_3 = 0.594$, $S_1 = 4.4$, $S_2 = 1.1$, $S_3 = 80$

$r_{12} = 0.80$, $r_{23} = -0.56$, $r_{13} = -0.40$.

(a) Find the correlation coefficient $r_{23.1}$ and $r_{1.23}$.

(b) Also estimate the value of x_1, when $x_2 = 6.0$ and $x_3 = 650$.

Solution : ৪৮০ পৃষ্ঠার **Problem-15** নং প্রশ্নের উত্তর দেখুন।

গ্রন্থ সহায়ক : উপরিউক্ত সাজেশন্সের পূর্ণাঙ্গ উত্তরপত্র পেতে.....
ব্যতিক্রম মার্কেটিং ইজিপ্রেস (২য় খণ্ড) বইটি সংগ্রহ করুন।

জাতীয় বিশ্ববিদ্যালয়ের প্রশ্নাবলি ও উত্তরপত্র

Business Statistics II (In English); Subject Code : 232307

National University

[BBA (Hon's) 3rd Year Exam-2019; (Held on-12.02.2020)]

(Department of Marketing)

Subject Code : 232307

Subject Name : Business Statistics-II (In English)

Time : 4 hours; **Full marks :** 80

[**N.B.** Answer different questions of each group chronologically.]

Group-A

1. Answer any **ten** questions. $1 \times 10 = 10$

(a) What is probability?

Ans. Probability is the favourable specific ratio of occurring or not occurring of any event.

(b) What is an experiment?

Ans. The activity that results in, or produces and event is regarded as experiment.

(c) What is sampling?

Ans. The methods for selecting. Sample from a Population is known in Sampling.

(d) What is mathematical expectation?

(e) What is population?

Ans. A population is the collection of all possible objects having certain characteristic under a study.

(f) What is normal distribution?

Ans. A distribution of a continuous random variable with a sing peaked bell shaped curve is called a standard normal distribution.

(g) What is hypothesis?

Ans. Hypothesis a statement about a Population Parameter developed for the Purpose of testing.

(h) What do you mean by type II error?

Ans. Type II error is the accepting a null hypothesis when it is false.

(i) What is chi-square distribution?

Ans. The distribution of the chi-square statistic is called the chi-square (χ^2) distribution.

(j) What is multiple correlation?

Ans. If we study the relationship among three or more variables then it is called multiple correlation.

(k) What is analysis of variance?

Ans. The Analysis of Variance is a technique that test for significant of the different among more than two sample means.

(l) What is binomial distribution?

Ans. The distribution which contains only two Possible outcomes (Such as Success or failure) is called binomial distribution.

Group-B

[Answer any **five** questions.] $4 \times 5 = 20$

2. Discuss the importance of probability in making business decision.

[Ans মার্কেটিং ম্যানেজমেন্ট (২য় খণ্ড) বইয়ের **Page No. 311; Ques. No. 1.26]**

3. What are the characteristics of poisson distribution? ***[Ans : Page No. 309; Ques. No. 1.17]***

4. You are given the following probability distribution for the annual sales for a company :

Sales (Tk. in crore)	Probability
200	0.10
250	0.20
275	0.40
310	0.25
350	0.50

Calculate the expected sale of the company. Also find the standard deviation of the sale.

Solution : ***Page No-353; Problem No-32***

5. Distinguish between census and sample. ***[Ans : Page No. 357; Ques. No. 3.15]***
6. Briefly describe the steps of testing hypothesis. ***[Ans : Page No. 381; Ques. No. 5.06]***
7. 800 people are involved in a factory. Out of them 350 are educated. In an efficiency test, 175 are terminated for their inefficiency and 45 of them are educated. Make a contingency table using the above data.
   Solution : ***Page No-425; Problem No-02***
8. What are the differences between null and alternative hypothesis? ***[Ans : Page No. 386; Ques. No. 5.23]***
9. Is the following information correct or consistent? **Solution :** ***Page No-469; Problem No-04***
   $r_{12} = 0.07, r_{13} = -0.6, r_{23} = 0.9$

Group-C

[Answer any **five** questions.] $5 \times 10 = 50$

10. A bag contains 6 red, 6 blue and 8 green balls. If 3 balls are drawn at random, determine the probability that:
    (i) all 3 are red
    (ii) all 3 are blue
    (iii) at least 1 is blue
    (iv) 2 are red and 1 is green
    (v) 1 of each color **Solution :** ***Page No-322 ; Problem No-15***
11. A population consists of the following values :
    2, 2, 4, 4 and 8.
    (i) List all samples of size 2 and compute the mean of each sample.
    (ii) Compute the mean of the distribution of sample means and the population mean. Compare the two values.
    (iii) Compare the dispersion in the population with that of the sample means.
    Solution : ***Page No-369; Problem No-10***
12. A population has three elements that are 4, 6, 8, Select the number of sample size of 2 and show that sample mean is riskless determinant to population mean. **Solution :** ***Page No-364; Problem No-06***
13. Two cards are drawn without replacement from a pack of 52 cards. Find the probability that the two cards are :
    (i) Same color Ace
    (ii) No king
    (iii) Different colors. **Solution :** ***Page No-327; Problem No-21***
14. Two types of new cars produced by a company are tested for petrol mileage. One group consisting of 36 cars averaged 14 kms per litre, while the other consisting of 72 cars averaged 12 kms per litre.
    (i) What test statistic is appropriate if :
    $S_1^2 = 1.5$ and $S_2^2 = 2.0$?
    (ii) Test whether there exists a significant difference in the petrol consumption of these two types of cars (use are 0.01). **Solution :** ***Page No-397; Problem No-13***
15. From the following data, test the significance of relation between education and skillness at 5% level of significance :

No. of labor

	Skilled	Un-skilled
Educated	75	20
Un-educated	25	30

Solution : ***Page No-429; Problem No-08***

16. By using the following information : **Solution :** ***Page No-475; Problem No-11***

Performance evaluation (X_1)	28	33	21	40	38	46
Aptitude Test Score (X_2)	74	87	69	69	81	97
Prior Experience (X_3)	5	11	4	9	7	10

    (i) Develop the estimating equation best describing these data.
    (ii) If an employee scored 84 on the aptitude test and had a prior experience of 7 years, what performance evaluation would be expected?
17. (a) What are the reasons of sample survey? ***[Ans : Page No. 356; Ques. No. 3.06]***
    (b) Explain the ways of reducing non sampling error. ***[Ans : Page No. 357; Ques. No. 3.14]***

National University

[BBA Hon's 3rd year; Exam-2020 (Held on-16/03/2022)]

(Department of Marketing)

Subject Code : 232307

Subject Name : Business Statistics-II (In English)

Time : 4 hours; **Full marks :** 80

[N.B. Answer different questions of each group chronologically.]

Group-A

1. Answer any **ten** questions. $1 \times 10 = 10$

(a) What is an event?

Ans. One or more of the possible outcomes of doing something, or one of the possible outcomes from conducting an experiment.

(b) What is parameter?

Ans. The values that desirable the characteristics of a Population are called Parameters.

(c) What is sample?

Ans. Sample is a portion of the elements of a Population Selected for direct measurement.

(d) What is permutation?

Ans. A permutation is an arrangement or ordering of a number of district objects. For example the words toped pot' represent two different partitions (or arrangements) of the same three letters.

(e) What is critical value?

Ans. The value of the standard statistic (Z or t) beyond which we reject the null hypothesis.

(f) What is census?

Ans. We can say the measurement or examination of every element in the population is called census.

(g) What is multiple regression?

Ans. When there are more than one independent variables then it is said multiple regression.

(h) What is random variable?

Ans. A variable that is allowed to take on any value within a given range.

(i) What is the probability distribution?

Ans. Probability distribution is a list of all possible outcomes of an experiment and the probabilities associated with each outcome.

(j) What is sampling error?

Ans. The error that arises due to drawing inference about a Population based on Sampling is known Sampling error.

(k) What is null hypothesis?

Ans. The hypothesis about a population parameter we wish to test is called a null hypothesis.

(l) What is degree of freedom?

Ans. The number of unrestricted chances for variation in the measurement being made.

Group-B

[Answer any **five** questions.] $4 \times 5 = 20$

2. Write down the differences between a prior and a posterior definitions of probability.
[Ans মার্কেটিং ম্যানেজমেন্ট (২য় খণ্ড) বইয়ের **Page No. 313; Ques. No. 1.32]**
3. What are the properties of a Binomial distribution? ***[Ans : Page No. 309; Ques. No. 1.15]***
4. Find the probability of getting 3 heads in tossing 5 coins. **Solution :** ***Page No-314; Problem No-02***
5. Distinguish between z-test and t-test. ***[Ans : Page No. 385; Ques. No. 5.20]***
6. Distinguish between sampling and non-sampling error. ***[Ans : Page No. 358; Ques. No. 3.18]***
7. What are the characteristics of normal distribution? Explain. ***[Ans : Page No. 308; Ques. No. 1.13]***
8. The mean and variance of a binomial distribution are 4 and $\frac{4}{3}$ respectively.

Find (i) probability function (ii) $P(x = 0)$, (iii) $P(x \geq 1)$. **Solution :** ***Page No-334; Problem No-07***
9. Heights in inches of 10 students are given below :
38, 40, 45, 53, 47, 43, 55, 48, 52, 49.
Test the variance of the distribution of height of all students from which the above sample of 10 students was drawn is equal to 20 square inches. **Solution :** ***Page No-434; Problem No-13***

Group-C

[Answer any **five** questions.] $5 \times 10 = 50$

10. A bag contains 6 red and 4 white balls. Two balls are drawn at random, What is the probability that :
    (i) All of them are red
    (ii) Same colour
    (iii) Different colour
    (iv) None of them is red. **Solution :** ***Page No-320 ; Problem No-13***

11. The income of a group of 10000 persons was found to be normally distributed with mean Tk. 1750 p.m. and standard deviation Tk. 50. Show that of this group 95% had income exceeding Tk. 1668 and only 5% had income exceeding Tk. 1832. What was the lowest income among the richest 100%.
    Solution : ***Page No-346; Problem No-20***

12. On the average, one in 400 items is defective if the items are packed in boxes of 100. What is the probability that any given box of items will contain :
    (i) No defective
    (ii) Less than two defectives
    (iii) One or more defectives
    (iv) More than three defectives. **Solution :** ***Page No-342; Problem No-17***

13. Intelligence test on groups of boys and girls gives the following result :

	Mean	**S.D.**	**Number**
Boys	70	20	250
Girls	75	15	150

Is there any significant difference in the mean scores obtained by boys and girls?
Solution : ***Page No-393; Problem No-07***

14. From the following data test the significance of relation between smoking and drinking :

	Drinking	**Not-drinking**
Smoking	18	26
Not-smoking	10	30

Solution : ***Page No-428; Problem No-06***

15. By using the following information, find the regression equation of X_1 on X_2 and X_3 :

X_1	12	22	32	28
X_2	6	12	16	22
X_3	4	6	12	18

Solution : ***Page No-473; Problem No-10***

16. Random samples drawn from two places gave the following data relating to the heights of adults males :

	Barishal	**Khulna**
No. of adults male in sample	1200	1500
Average height (in inches)	68.50	68.58
Standard deviation (inches)	2.5	3.0

Test at 5% level of significance, that the mean height is the same for adults males in two places.
Solution : ***Page No-395; Problem No-10***

17. The Drug International Company wishes to test whether its three salesmen X, Y and Z tend to make sales of the same size or whether they differ in their selling ability as measured by the average size of their sales. During the last week, out of 14 sales, X made 5, Y made 4 and Z made 5 calls. The following are the weekly sales (in Tk. thousand) record of three salesmen :

X	**Y**	**Z**
300	600	700
400	300	300
300	300	400
500	400	600
0	---	500

Test whether the three salesmen's average sales differ in size. **Solution :** ***Page No-414; Problem No-04***

National University

[BBA Hon's 3rd year; Exam-2021 (Held on-04/06/2023)]

(Department of Marketing)

Subject Code : 232307

Subject Name : Business Statistics-II (In English)

Time : 4 Hours **Full Marks :** 80

[N.B. Answer different questions of each group chronologically.]

Group-A

1. Answer any **ten** questions. $1 \times 10 = 10$

(a) What is probability?

Ans. Probability is the favourable specific ratio of occurring or not occurring of any event.

(b) What is an experiment?

Ans. The activity that results in, or produces and event is regarded as experiment.

(c) What is binomial distribution?

Ans. The distribution which contains only two Possible outcomes (Such as Success or failure) is called binomial distribution.

(d) What is sampling?

Ans. The methods for selecting. Sample from a population is known in sampling.

(e) What is type-II error?

Ans. Type II error is the accepting a null hypothesis when it is false.

(f) What is hypothesis?

Ans. Hypothesis a statement about a population parameter developed for the purpose of testing.

(g) What is analysis of variance?

Ans. The analysis of variance is a technique that test for significant of the different among more than two sample means.

(h) What is test of hypothesis?

Ans. Test of hypothesis is a statistical Procedure that is used to provide evidence in favour of same statement.

(i) What is partial correlation?

Ans. Partial correlation is measure of the strength and direction of a linear relationship between two continuous variables whilst controlling for the effect of one or more other continuous variables (also known as covariates or control variables).

(j) Who has developed t-test?

Ans. William Sealy Gosset developed t-test.

(k) What do you mean by chi-square distribution?

Ans. The distribution of the chi-square statistic is called the chi-square (χ^2) distribution.

(l) What is population?

Ans. A population is the collection of all possible objects having certain characteristic under a study.

Group-B

[Answer any **five** questions.] $4 \times 5 = 20$

2. Discuss the importance of probability in making business decision.

[Ans : [illegible] মার্কেটিং ইন্টারেস্ট (২য় খণ্ড) বইয়ের Page No. 311; Ques. No. 1.26]

3. Distinguish between type-I and type-II error. ***[Ans : Page No. 386; Ques. No. 5.24]***

4. What are the characteristics of poisson distribution? ***[Ans : Page No. 309; Ques. No. 1.17]***

5. Distinguish between census and sample. ***[Ans : Page No. 357; Ques. No. 3.15]***

6. Discuss the properties of chi-square test. ***[Ans : Page No. 422; Ques. No. 7.05]***

7. An unbiased coin is tossed four times. What is the probability of gettting at least 2 heads?

Solution : ***Page No-315; Problem No-03***

8. 800 people are involved in a factory. Out of them 350 are educated. In an efficiency test, 175 are terminated for their inefficiency and 45 of them are educated. Make a contingency table using the above data.

Solution : ***Page No-425; Problem No-02***

মার্কেটিং (অনার্স-৩য় বর্ষ)-২২(ক)

9. The following zero-order, correlation coefficients are given :
$r_{12} = 0.98$, $r_{13} = 0.44$ and $r_{23} = 0.54$
Calculate multiple correlation coefficient treating first variable as dependent and second and third variables as independent. **Solution :** ***Page No-468; Problem No-02***

Group-C

[Answer any **five** questions.] $5 \times 10 = 50$

10. A box contains 6 red, 7 blue and 8 green balls. If 3 balls are drawn at random, determine the probability that :
(i) All 3 are red
(ii) All 3 are blue
(iii) At least 1 is blue
(iv) 2 are red and 1 is green
(v) 1 of each color. **Solution :** ***Page No-324 ; Problem No-16***

11. The probability that a patient survives from neurosurgery is 0.70. If 10 patients have had the surgery, then what is the probability that:
(i) None will survive
(ii) At least one will survive
(iii) At most 3 will survive
(iv) Between 2 and 4 will survive.
Solution : ***Page No-336; Problem No-09***

12. A population consists of the following values :
2, 2, 4, 4 and 8.
(i) List all sample of size 2 and compute the mean of each sample.
(ii) Compute the mean of the distribution of sample means and the population mean. Compare the two values.
Solution : ***Page No-369; Problem No-10***

13. Speed Unlimited motors Ltd. had 5 representatives, the number of sold cars in the last month are given :

Representative	A	B	C	D	E
Sold car	5	15	10	25	20

14. Mean and standard deviation of production of the two worker groups are given below :

	1st Group	**2nd Group**
Number at workers	34	20
Mean of production	50 tons	55 tons
Standard deviation	6 tons	5 tons

Test the significant differences between the two groups.
Solution : ***Page No-405; Problem No-19***

15. Two cards are drawn without replacement from a pack of 52 cards. Find the probability that the two cards are :
(i) Same color Ace
(ii) No king
(iii) Different colors.
Solution : ***Page No-327; Problem No-21***

16. From the following data, test the significance of relation between education and skillness at 5% level of significance :

No. of Labor

	Skilled	Un-skilled
Educated	75	20
Un-educated	25	30

Solution : ***Page No-429; Problem No-08***

17. (a) Discuss the reasons of sample survey. ***[Ans : Page No. 356; Ques. No. 3.06]***
(b) Explain the ways of reducing non sampling error. ***[Ans : Page No. 357; Ques. No. 3.14]***

National University

[BBA (Hon's) 3rd Year; Exam-2022 (Held in-06/05/2024)]

(Department of Marketing)

Subject Code : 232307

Subject Name : Business Statistics-II (In English)

Time : 4 hours; **Full marks :** 80

[**N.B.** Answer different questions of each group chronologically.]

Group-A

1. Answer any **ten** questions. $1 \times 10 = 10$

(a) What is business statistics?

Ans : Business statistics is a science assisting us to make business decision under uncertainties based on some numerical and measurable scale.

(b) What is parameter?

Ans : The values that desirable the characteristics of a population are called parameters.

(c) What is the formula of poisson distribution?

Ans : The formula of poisson distribution $P(x) = \frac{e^{-m} m^x}{x!}$

(d) What is critical value?

Ans : The value of the standard statistic (Z or t) beyond which we reject the null hypothesis.

(e) What is null hypothesis?

Ans : The hypothesis about a population parameter we wish to test is called a null hypothesis.

(f) What is statistical estimation?

Ans : Statistical estimation is the process of using a sample statistic to estimate a population parameter.

(g) What is the meaning of P(A) = 1?

Ans : Meaning of P(A) = 1, the total number of elements of event A and sample space are equal. That is implies A is a certain event.

(h) What is small sample test?

Ans : When sample size become at least 30 or not more then 30 then used method in statistical test of hypothesis is called small sample test.

(i) What is multiple regression?

Ans : When there are more than one independent variables then it is said multiple regression.

(j) Write the formula of chi-square test?

Ans : $\chi^2 = \sum \frac{(O - E)^2}{E}$

(k) What is the full meaning of ANOVA?

Ans : Analysis of Variance.

(l) An unbiased coin is tossed three times. Write down the sample space.

Ans : Determination of sample space :

Total sample are $= 2^n = 2^3 = 8$

2 coin / 1 coin	HH	HT	TH	TT
H	HHH	HHT	HTH	HTT
T	THH	THT	TTH	TTT

$\therefore$ Sample space, s = {HHH, HHT, HTH, HTT, THH, THT, TTH, TTT}

Group-B

[Answer any **five** questions.] $4 \times 5 = 20$

2. A card is drawn from a pack of 52 cards. Find the probability of getting (i) King (ii) Club.

Solution : মার্কেটিং ইনভেন্টরি (২য় খণ্ড) বইয়ের **Page No-326 ; Problem No-18**

3. Discuss the properties of binomial distribution. ***[Ans : Page No. 309; Ques. No. 1.15]***

4. Write down the differences between sampling and non sampling error. ***[Ans : Page No. 358; Ques. No. 3.18]***

5. Assume that on an average one telephone number out of 15 is busy. What is the probability that if six randomly selected telephone numbers are called not more than three will be busy? **Solution :** ***Page No-335; Problem No-08***

6. In a survey of 100 readers, it was found 40 read the Daily Star and 15 read The Daily Sun and 10 read both. What is the probability of a person reading at least one of the newspapers? **Solution :** ***Page No-350; Problem No-26***

7: The mean of a binomial distribution is 40 and standard deviation 6. Calculate N, P and Q.
Solution : ***Page No-332; Problem No-04***

8. Distinguish between z-test and t-test. ***[Ans : Page No. 385; Ques. No. 5.20]***

9. If $r_{12} = 0.8$, $r_{13} = -0.4$ and $r_{23} = -0.56$.
Find the value of $r_{12.3}$. **Solution :** ***Page No-469; Problem No-05***

Group-C

[Answer any **five** questions.] $10 \times 5 = 50$

10. On the average, one in 400 items is defective if the items are packed in boxes of 100. What is the probability that any given box of items will contain :
(i) No defectives
(i) Less than two defectives
(iii) Exactly three defectives. **Solution :** ***Page No-342 ; Problem No-17***

11. In a community the probability that a newly born child will be a girl is 0.4. Among 4 newly born children in that community, determine the probability that :
(i) All the four girls
(ii) At least two girls
(iii) No girl
(iv) Exactly one girl
(v) At most two girls. **Solution :** ***Page No-337; Problem No-11***

12. Three unbaised coins are tossed at a time.
(a) Write down the sample space.
(b) Calculate the probabilities of the following :
(i) 1 head
(ii) 1 tail
(iii) More than 1 head
(iv) At best 2 heads
(v) At least 2 heads. **Solution :** ***Page No-315; Problem No-04***

13. The income of a group of 10000 persons was found to be normally distributed with mean Tk. 1750 per day and standard deviation Tk. 50. Show that of this group 95% had income exceeding Tk. 1668 and only 5% had income exceeding Tk. 1832. What was the lowest income among the richest 100%? **Solution :** ***Page No-346; Problem No-20***

14. You are working as a purchase manager for a company. The following information has been supplied to you by two manufacturers of electric bulbs.

	Company-A	Company-B
Mean life	1300 hours	1288 hours
Standard deviation	82 hours	93 hours
Sample size	80	80

Which brand of bulbs are you going to purchase if you desire to take a risk of 5%? **Solution :** ***Page No-396; Problem No-11***

15. The following table gives the numbers of good and bad parts produced by each of three shifts in a factory.

Shift	Good	Bad
Day	900	130
Evening	700	170
Night	400	200

If there any association between the shift and equality of parts produced? **Solution :** ***Page No-430; Problem No-09***

16. From the following data, test the significance of relation between facebook user and age at 5% level of significance : **Solution :** ***Page No-426; Problem No-04***

	Young	Old
Facebook user	18	28
Facebook nonuser	15	35

17. Draw a sample of 3 units from a population of (4, 10, 3, 7, 8) Find —
(i) Write down the samples
(ii) Mean of sample means
(iii) Sample mean and population mean
(iv) Showing that mean of sample mean is equal to population mean. **Solution :** ***Page No-366; Problem No-08***

National University

[BBA (Hon's) 3rd Year; Exam-2023 (Held in-04/05/2025)]

(Department of Marketing)

Subject Code : 232307

Subject : Business Statistics-II (In English)

Time : 4 hours; **Full marks :** 80

[N.B. Answer different questions of each group chronologically.]

Group-A

Answer any **ten** questions. Marks : 1 × 10 = 10

(a) What is probability?

Ans. Probability is the favorable specific ratio of occurring or not occurring of any event.

(b) What is population?

Ans. A population is the collection of all possible objects having certain characteristic under a study.

(c) What is sampling?

Ans. The methods for selecting sample from a population is known as sampling.

(d) What is hypothesis?

Ans. Hypothesis a statement about a population parameter developed for the purpose of testing.

(e) What is analysis of variance?

Ans. The analysis of variance is a technique that test for significant of the different among more than two sample means.

(f) What is partial correlation?

Ans. Partial correlation is measure of the strength and direction of a linear relationship between two continuous variables whilst controlling for the effect of one or more other continuous variables (also known as covariates or control variables).

(g) What is sampling error?

Ans. The error that arises due to drawing inference about a Population based on Sampling is known Sampling error.

(h) What is normal curve?

Ans. The probability graph of a normal distribution is called the normal curve.

(i) What is parametric test?

Ans. A parametric test is a type of statistical test that makes specific assumptions about the parameters of the population distribution from which the sample is drawn.

(j) What is random variable?

Ans. A variable that is allowed to take on any value within a given range.

(k) What is census survey?

Ans. We can say the measurement or examination of every element in the population is called census.

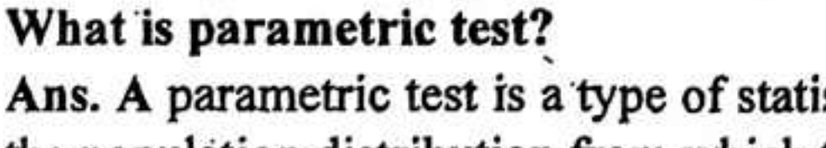
(l) What is Poisson distribution?

Ans. A discrete probability distribution in which the probability of occurrence of an outcome within a wary small, and the probability that two or more such outcomes will occur within the same small time interval is negligible. The occurrence of an outcome within on time period is independent of the other.

Group-B

[Answer any **five** questions.] Marks : 4 × 5 = 20

2. Distinguish between type-I and type-II error.

[Ans :  **Page No. 386; Ques. No. 5.24]**

3 What are the characteristics of Poisson distribution? ***[Ans : Page No. 309; Ques. No. 1.17]***

4. In a Poisson distribution, $P(x = 2) = P(x = 3)$. Find the probability function.

Solution : ***Page No-340; Problem No-14***

5. Discuss the uses of normal distribution. ***[Ans : Page No. 312; Ques. No. 1.29]***

6. Distinguish between census and sample. ***[Ans : Page No. 357; Ques. No. 3.15]***

7. A hawker earn Tk. 2000, if it is fair day. If it rains, he losses Tk. 250. If the probability of fair day is 0.6, then find his expected profit or loss per day. **Solution :** ***Page No-353; Problem No-31***

8. The mean and variance of a binomial distribution are 4 and $\frac{4}{3}$ respectively.

   Find : (i) the probability function and (ii) P (X = 0). **Solution :** ***Page No-334; Problem No-07***

9. Show the differences between a prior and a posterior definitions of probability. ***[Ans : Page No. 313; Ques. No. 1.32]***

Group-C

[Answer any **five** questions.] Marks : 10 × 5 = 50

10. A bag contains 6 red and 4 white balls. Two balls are drawn at random, What is the probability that :
    (i) All of them are red
    (ii) Same colour,
    (iii) Different colour
    (iv) None of them is red. **Solution :** ***Page No-320 ; Problem No-13***

11. The probability that a patient survives from neurosurgery is 0.70. If 10 patients have had the surgery, then what is the probability that :
    (i) None will survive.
    (ii) At least one will survive.
    (iii) At most three will survive.
    (iv) Between two and four will survive. **Solution :** ***Page No-336; Problem No-09***

12. The amount of money requested on loan applications at 'Pubali Bank' follow the normal distribution, with a mean of Tk. 70,000 and standard deviation of Tk. 20,000. A loan application is recorded this morning. Find the probability that :
    (i) The amount requested is Tk. 80,000 or more.
    (ii) The amount requested is between Tk. 65,000 and Tk. 80,000. **Solution :** ***Page No-345; Problem No-19***

13. Intelligence test on two groups of boys and girls gives the following result :-

	Mean	Standard deviation	Numbers
Girls	90	10	50
Boys	84	12	70

Determine 95% confidence interval of the difference in the mean scores of boys and girls?

Solution : ***Page No-394; Problem No-08***

14. A random sample of 100 students gave a mean weight of 58 kg with a standard deviation of 4 kg. Test the hypothesis that the mean weight in the population is 60 kg. **Solution :** ***Page No-391; Problem No-02***

15. From the following data, test the significance of relation between education and skillness at 5% level of significance : **Solution :** ***Page No-429; Problem No-08***

	Skilled	Un-skilled
Educated	75	20
Un-educated	25	30

16. By using the following information, find the regression equation of X_1 on X_2 and X_3 :

Solution : ***Page No-473; Problem No-10***

X_1	12	22	32	28
X_2	6	12	16	22
X_3	4	6	12	18

17. (a) Discuss the reasons of sample survey. ***[Ans : Page No. 356; Ques. No. 3.06]***
    (b) Explain the ways of reducing non-sampling error. ***[Ans : Page No. 357; Ques. No. 3.14]***

National University

[BBA Hon's 3rd Year; Exam-2024 (Held on- 07/01/2026)]

(Department of Marketing)

Subject Code : 232307

Subject Name : Business Statistics-II (In English)

Time : 4 hours; **Full marks :** 80

[**N.B.** Different parts of each question must be written consecutively.]

Part-A

1. Answer any **ten** questions : $1 \times 10 = 10$

(a) Define joint probability.

Ans : The probability of two events occurring together or in succession is called joint probability.

(b) What is continuous random variable?

Ans : A variable that is allowed to take on any value within a given range is called a continuous random variable.

(c) What is t-distribution?

Ans : T-distribution is a probability distribution of the standardized sample mean when the population variance is unknown and the sample size is small.

(d) What is contingency table?

Ans : The cross classification can be conveniently displayed means of a table is called contingency table.

(e) Define Chi-square distribution.

Ans : The distribution of the chi-square statistic is called the chi-square (χ^2) distribution.

(f) What is an experiment?

Ans : The activity that results in, or produces and event is regarded as an experiment.

(g) What is the probability distribution?

Ans : Probability distribution is a list of all possible outcomes of an experiment and the probabilities associated with each outcome.

(h) What is parameter?

Ans : The values that desirable the characteristics of a population are called parameters.

(i) What is degree of freedom?

Ans : The number of unrestricted chances for variation in the measurement being made is called the degree of freedom.

(j) What is null hypothesis?

Ans : The hypothesis about a population parameter we wish to test is called a null hypothesis.

(k) What is the full meaning of ANOVA?

Ans : Analysis of Variance.

(l) What is multiple regression?

Ans : When there is more than one independent variables then it is called multiple regression.

Part-B

[Answer any **five** questions.] Marks : $4 \times 5 = 20$

2. What are the properties of binomial distribution?

[Ans মার্কেটিং ইনভেস্টিগেশন (২য় খণ্ড) বইয়ের **Page No. 309; Ques. No. 1.15]**

3. A sample of 81 observations is taken from a normal population. The sample mean is 40 and the sample standard deviation is 5. Determine the 95% confidence interval for the population mean.

Solution : ***Page No-363; Problem No-04***

4. The probability of a man to be smoker is $\frac{1}{3}$. What is the probability of 2 men among 5 men to be smokers?

Solution : ***Page No-544; Question No-04***

5. If $r_{12} = 0.86$; $r_{13} = 0.65$ and $r_{23} = 0.72$; find the partial correlation co-efficient of $r_{12.3}$.

Solution : ***Page No-544; Question No-05***

6. An unbiased coin is tossed four times. What is the probability of getting at least 2 heads?

Solution : ***Page No-315; Problem No-03***

7. Distinguish between sampling and non sampling error. ***[Ans : Page No. 358; Ques. No. 3.18]***

8. The mean of binomial distribution is 40 and standard deviation is 6. Calculate N, P and Q.

Solution : ***Page No-332; Problem No-04***

9. Distinguish between z-test and t-test. ***[Ans : Page No. 385; Ques. No. 5.20]***

Part-C

[Answer any **five** questions.] Marks : 10 × 5 = 50

10. Describe the sampling methods. ***[Ans : Page No. 359; Ques. No. 3.19]***

11. A telephone operator receives 3 telephone calls on average per minute from 10 a.m. to 12 p.m. Find the probability that in a certain minute of the given time interval of a day, the operator receives :
    (i) No call
    (ii) At least two calls
    (iii) At best two calls
    (iv) Two or three calls.

    Solution : ***Page No-341; Problem No-16***

12. In a recent interview on banking profession, 32 out of 150 male banking professionals rated the banking career as 'Excellent' and 17 out of 80 female banking professional rated it as 'Excellent'. Using 10% level of significance, can we conclude that there is a significant difference in the proportion of male versus female banking professionals who rate the banking career as excellent?

    Solution : ***Page No-545; Question No-12***

13. To study the performance of three detergents and three different temperatures, the following 'Whiteness' readings were obtained with specially designed equipment :

Water temperature	Detergent		
	A	B	C
Cold	57	55	67
Warm	49	52	68
Hot	54	46	58

    Using 5% level of significances, perform a two-way analysis of variance.

    Solution : ***Page No-547; Question No-13***

14. On average, one in 400 items is defective. If the items are packed in boxes of 100. What is the probability that any given box of items will contain :
    (i) No defective
    (ii) Less than two defectives
    (iii) One or more defectives
    (iv) More than three defectives.

    Solution : ***Page No-342; Problem No-17***

15. Draw a sample of 3 units from a population of (4, 10, 3, 7, 8) and find :
    (i) Write down the samples.
    (ii) Obtain sample mean and population mean.
    (iii) Show that mean of sample mean is equal to population mean.

    Solution : ***Page No-366; Problem No-08***

16. Mean and standard deviation of production of the two worker groups are given below :

	1st group	2nd group
Number of workers	34	20
Mean of protluction	50 tons	55 tons
Standard deviation	6 tons	5 tons

    Is there any significant difference between the groups at 5% level of significance?

    Solution : ***Page No-405; Problem No-19***

17. Four salesmen were posted in different areas by a company. The number of units sold by them is given :

A	20	23	28	29
B	25	32	30	21
C	23	28	35	18
D	15	21	19	25

    From the above information, can it be concluded that there is a significant difference in the performance of salesmen at 5% level of significance?

    Solution : ***Page No-455; Problem No-06***